

Optimizing the Energy Profiles of a Building's Apartments to Support Decision-Making in Demand Response Programs

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Problem definition:

- The explosion in the number of electrical devices places a **great stress on the electrical grid**
- At peak hours there is a **high risk of electricity fallout** as a result of a collapse of the distribution network
- **Residential users** consume around **30-40%** of the total energy produced in the world ^[1]
- **Demand response programs:** Incentivized programs that **involve consumers** to **reduce the stress** from the grid at peak hours

Objectives:

- **Objective 1:**
 - Shift consumption from peak to off-peak hours
 - **Make a schedule** for appliances such that a **target consumption curve is approximated**, and the **discomfort** created is **minimized**
- **Objective 2:**
 - Select a **minimum subset of apartments** from a residential building that will have to modify their consumption pattern
 - **Schedule appliances** from the selected apartments to **approximate a target consumption curve** while **minimizing the discomfort**

Related Work – Load Scheduling

Reference	Objective	Meta-heuristic-based	Suitable for old buildings	Is comfort considered?	Number of classes used to classify appliances (device types)
[3]	cost reduction based on day-ahead prediction	X		YES	2
[4]	cost reduction based on day-ahead prediction			YES	1
[5]	Cost reduction	X	X	YES	1
[6]	Cost reduction		X	NO	3
[7]	Cost reduction	X		YES	4
[8]	Cost reduction	X	X	NO	3
[9]	Cost reduction	X	X	YES	4
[10]	Cost reduction	X	X	YES	4
My approach	Curve approximation	X	X	YES	2

Proposed Scheduling method

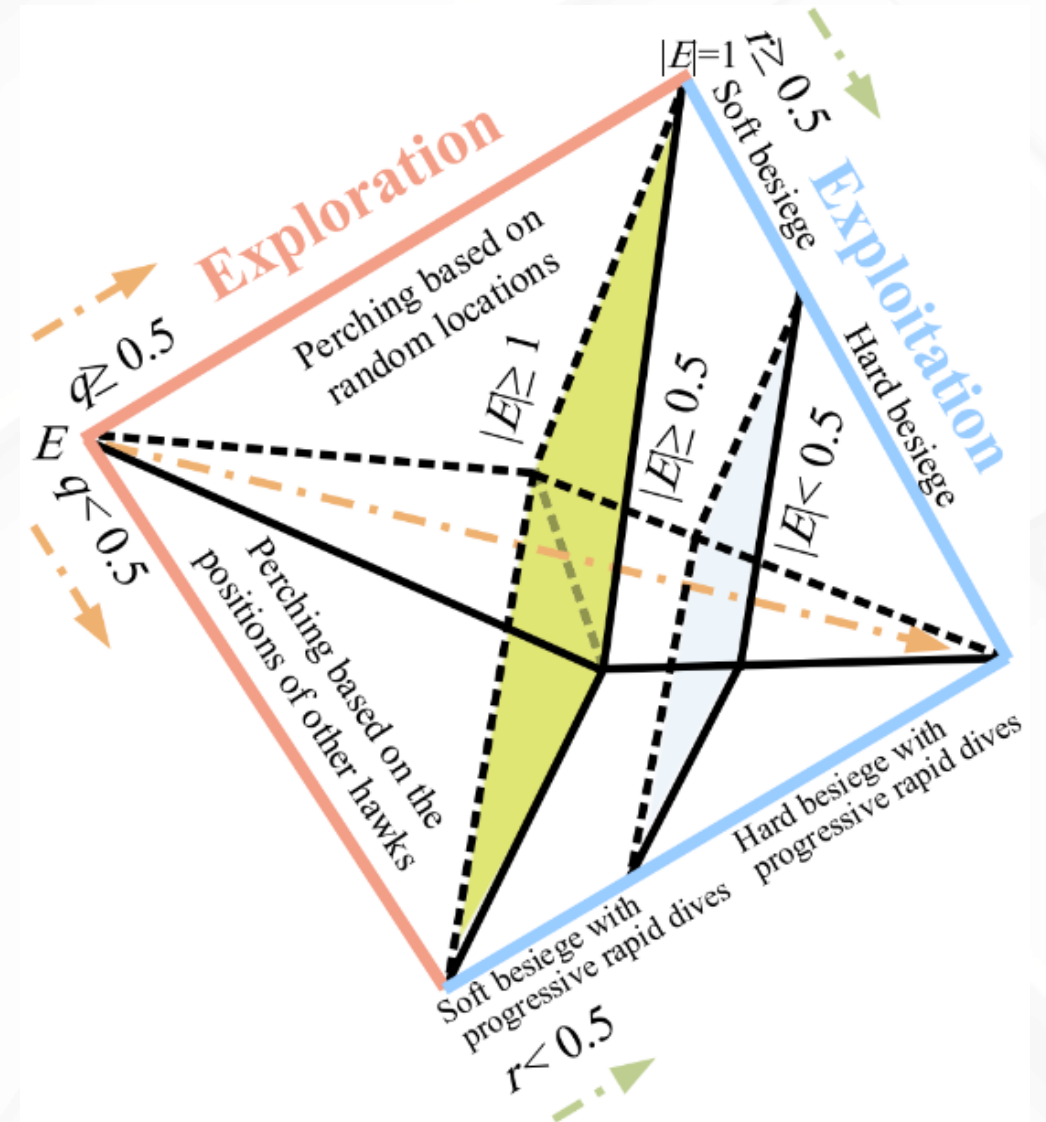
General assumptions

- A residential **building** contains apartments:
 - $B = \{ap_a \mid a \in [1, AN]\}$, AN – number of apartments
- Each **apartment** has an identifier and several devices:
 - $ap_a = (id, D_a)$ $D_a = \{d_{i_a} \mid i \in [1, DN_a]\}$, DN – number of devices in apartment a
- Each **device** has the following information defined:
 - id, name
 - **Consumption** (kWh)
 - **Constraints**: isDeferrable, minUsageHours
 - **Initial usage vector**: $S_{d_a} = \{isUsed_h \mid h \in \{0, 1, \dots, 23\}\}$
 - **Comfort vector**: $CV_{d_a} = \{discomfortRating_h \mid h \in \{0, 1, \dots, 23\}\}$
- **Target consumption curve**: $TC = \{consumption_h \mid h \in \{0, 1, \dots, 23\}\}$

Proposed Scheduling method

Objective 1 – Harris Hawks optimization

- inspired by the **haunting behavior of Harris Hawks**
- balancing **exploration and exploitation** based on an energy variable
- population **initialized randomly** onto the search space
- each search agent contains a schedule variant for the whole day of the building



Proposed Scheduling method

Objective 1 – Internal data representation

- The way of representing data in the optimization process:

id	consumption
(1,1)	0.025
(1,2)	0.0185
(1,3)	0.013
(2,1)	0.027
(2,2)	0.013
(3,1)	0.027
(3,2)	1.8
(3,3)	0.035

Figure 2. Consumption vector

id	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
(1,1)	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
(1,2)	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
(1,3)	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0
(2,1)	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	1	1	1	1	1	0	0	0
(2,2)	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0
(3,1)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(3,2)	1	1	0	1	1	0	0	1	1	1	0	0	0	0	0	1	0	1	0	1	1	1	0	1
(3,3)	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0

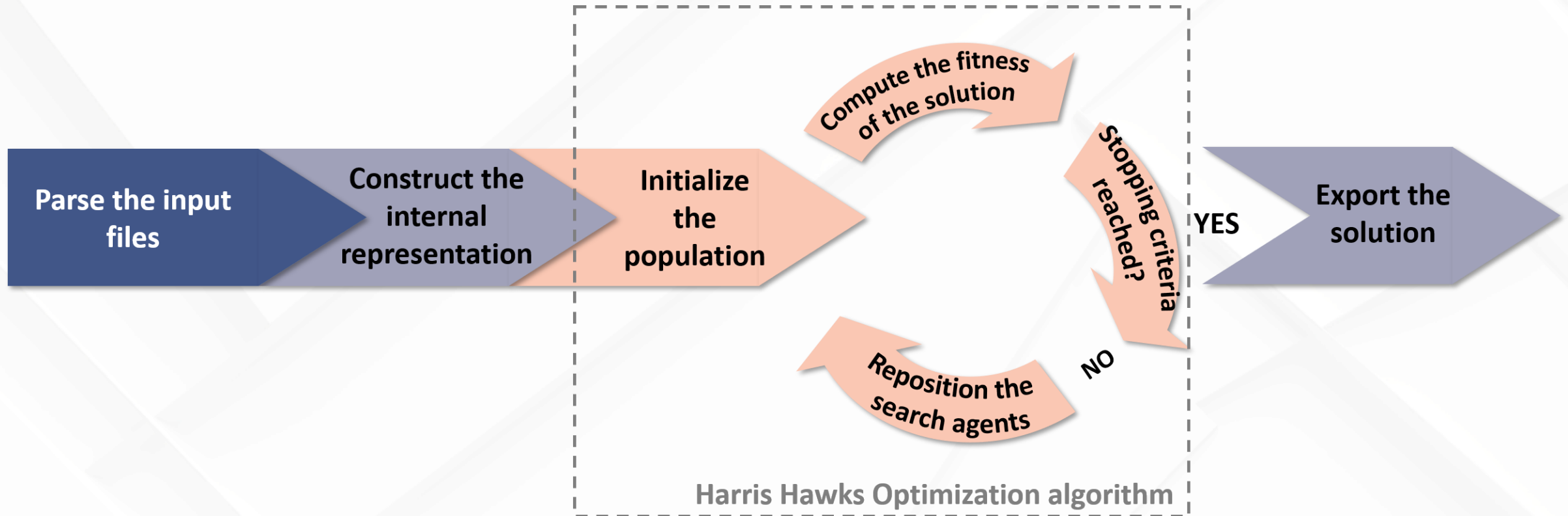
Figure 3. Usage matrix

id	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
(1,1)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(1,2)	0	0	2	1	0	0	0	1	2	2	2	2	2	2	2	2	2	2	2	1	0	0	0	0
(1,3)	2	2	2	2	2	2	0	1	0	2	2	2	2	2	2	2	2	2	0	1	1	0	0	1
(2,1)	2	2	2	2	2	2	2	2	2	2	1	0	1	1	0	0	1	1	0	0	0	1	2	2
(2,2)	2	2	2	2	2	2	0	0	0	2	2	2	2	2	2	2	2	2	0	2	2	0	0	1
(3,1)	2	2	2	2	2	2	2	2	2	2	0	0	0	1	0	1	0	0	0	1	2	2	2	2
(3,2)	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0
(3,3)	2	2	2	2	2	2	2	2	1	0	0	0	1	1	2	0	0	0	0	1	2	2	2	2

Figure 4. Comfort matrix

Proposed Scheduling method

Objective 1 – Optimization process (approximation and comfort)



Proposed Scheduling method

Objective 1 – Fitness function

$$f_{obj_lower} = (1 - w) * NF_{Approx} + w * NF_{comfort}$$

$$NF_{Approx} = w_{Euclidean} * NF_{Euclidean} + (1 - w_{Euclidean}) * NF_{Pearson}$$

$$NF_{Euclidean} = \left(\sum_{h=0}^{23} |SC[h] - TC[h]| \right) * \frac{100}{baseline_value}$$

$$NF_{Pearson} = \left(\frac{\sum_{h=0}^{23} (SC[h] - \overline{SC})(TC[h] - \overline{TC})}{\sqrt{\sum_{h=0}^{23} (SC[h] - \overline{SC})^2} \sqrt{\sum_{h=0}^{23} (TC[h] - \overline{TC})^2}} \right) * \frac{100}{baseline_value}$$

$$NF_{comfort} = \left(\sum_{h=0}^{23} \sum_{a=0}^{AN} \sum_{d=0}^{DN_a} P^{OCP_{d_a}}[h] - 1 \right) * \frac{100}{baseline_value}$$

w – comfort trade-off

$w_{Euclidean}$ = weight of the Euclidean distance evaluation function

SC – scheduled consumption

TC – target consumption

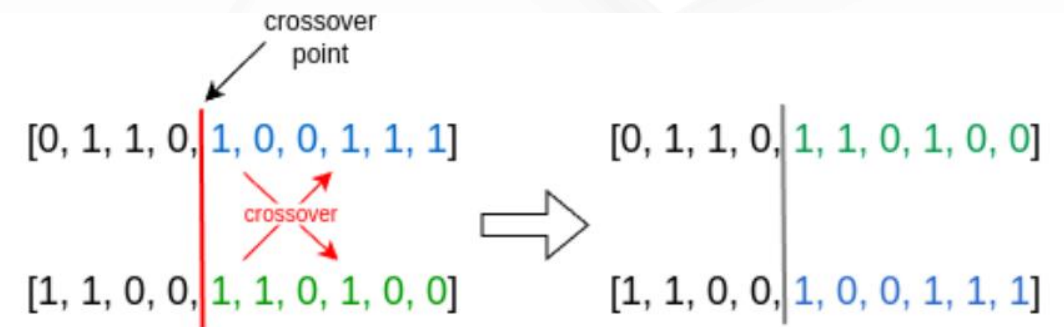
OCP – overall comfort penalization

P – comfort penalization base

Proposed Scheduling method

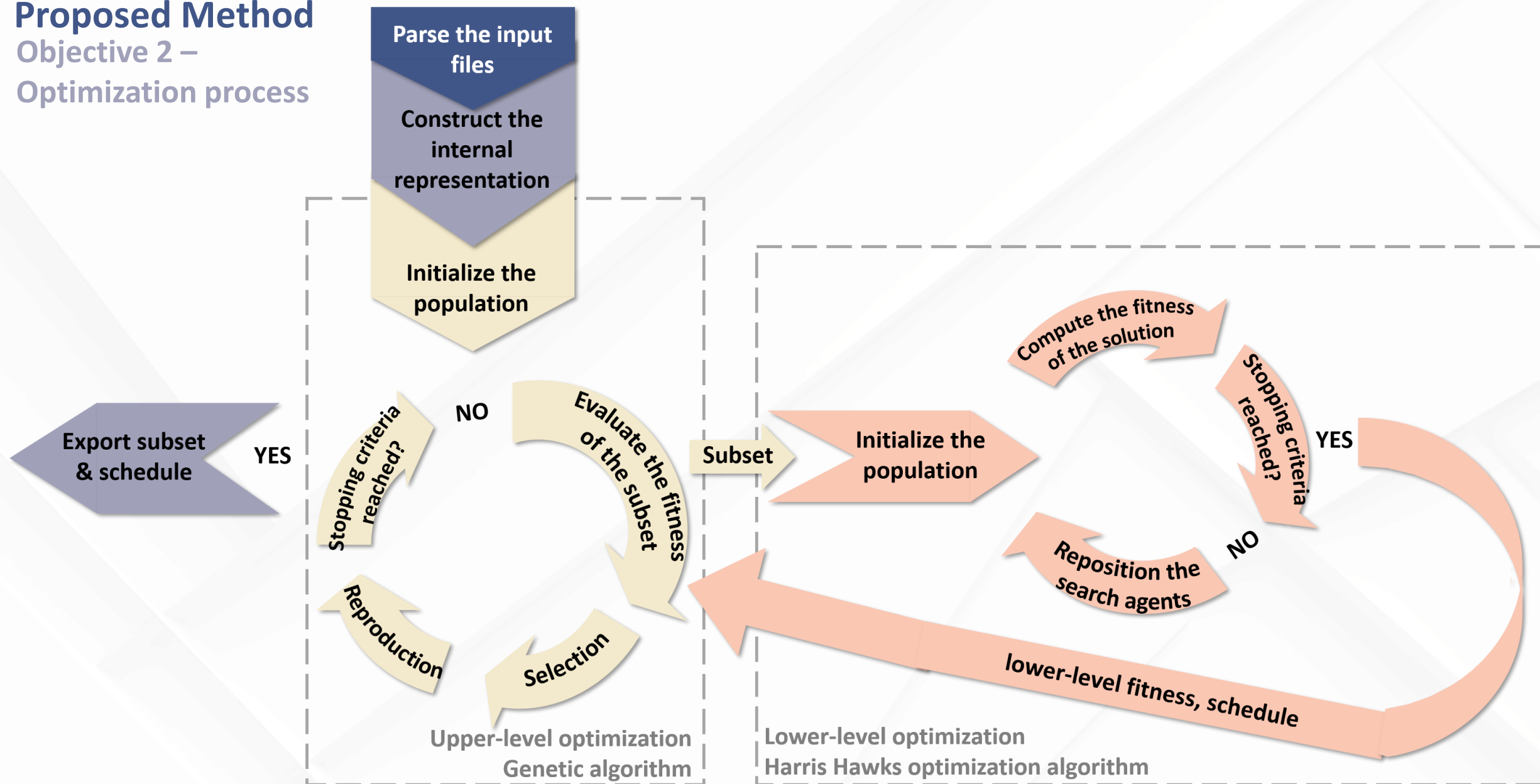
Objective 2 – Genetic algorithm

- inspired by the natural **process of selection and evolution**
- each agent contains a **Boolean vector** marking whether an **apartment** should or should not **alter its consumption** pattern
- **specific operations:**
 - selection – tournament
 - crossover – uniform
 - mutation – flip



Proposed Method

Objective 2 – Optimization process



Proposed Scheduling method

Objective 2 - Fitness function

$$f_{obj_up} = w_{subs} * F_{subs} + w_{over_sampling} * F_{over_sampling} + w_{approx_up} * NF_{Approx} + w_{comf_up} * NF_{comfort}$$

$$F_{subs} = \frac{AIN}{AN} * 100$$

$$F_{over_sampling} = \frac{HOC}{24} * 100$$

$$HOC = \sum_{h=0}^{23} isOverConsumption[h]$$

$$isOverConsumption = \begin{cases} 0, & TC[h] \geq CNI[h] \\ 1, & TC[h] < CNI[h] \end{cases}$$

w_{subs} – weight subset objective

$w_{over_sampling}$ – weight over-sampling objective

w_{approx_up} – weight approximation objective

w_{comf_up} – weight comfort objective

AIN – number of apartments included in subset

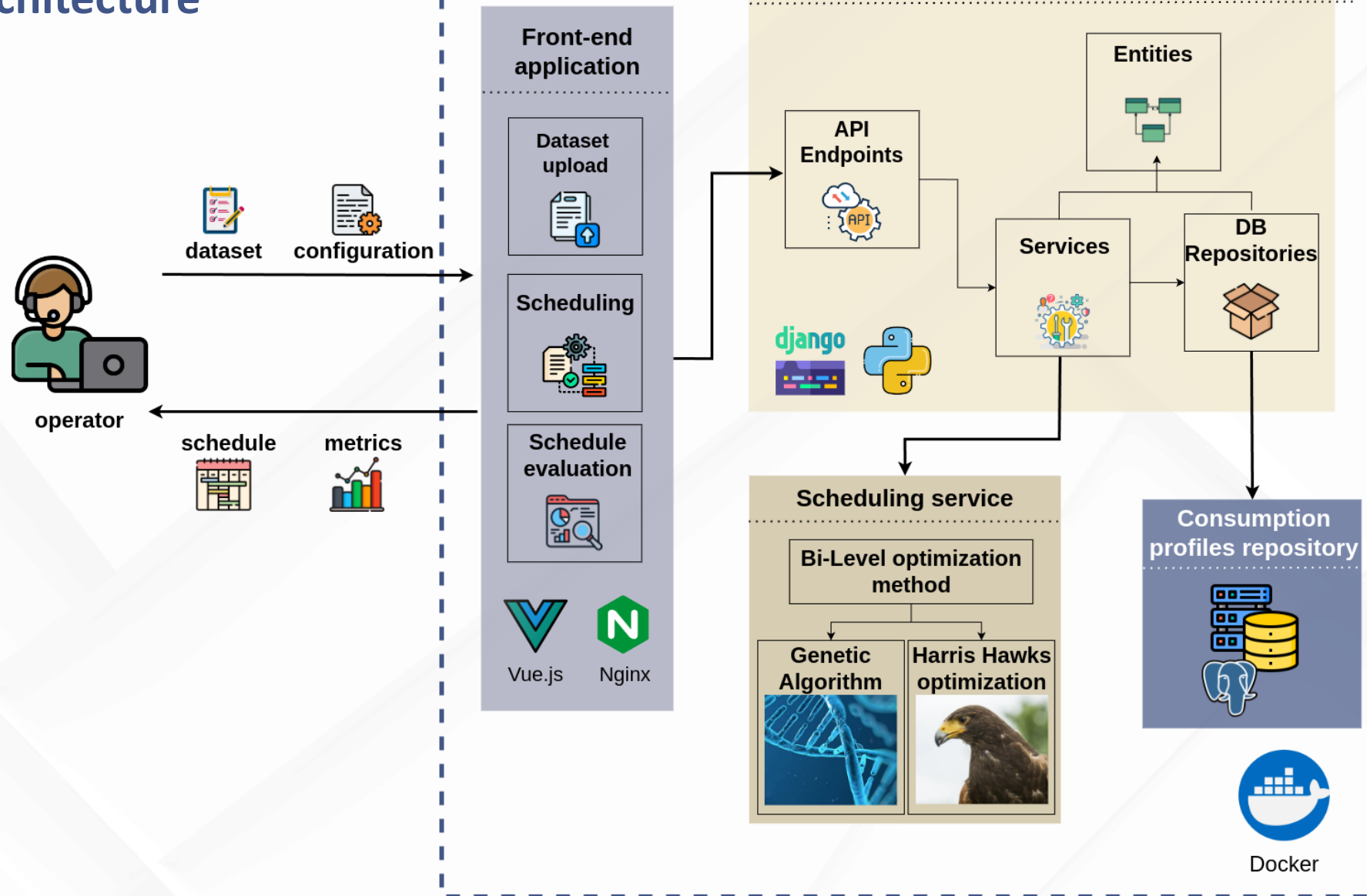
AN – total number of apartments

HOC – hours over consumption

TC – target consumption curve

CNI – consumption of apartments not included

System architecture

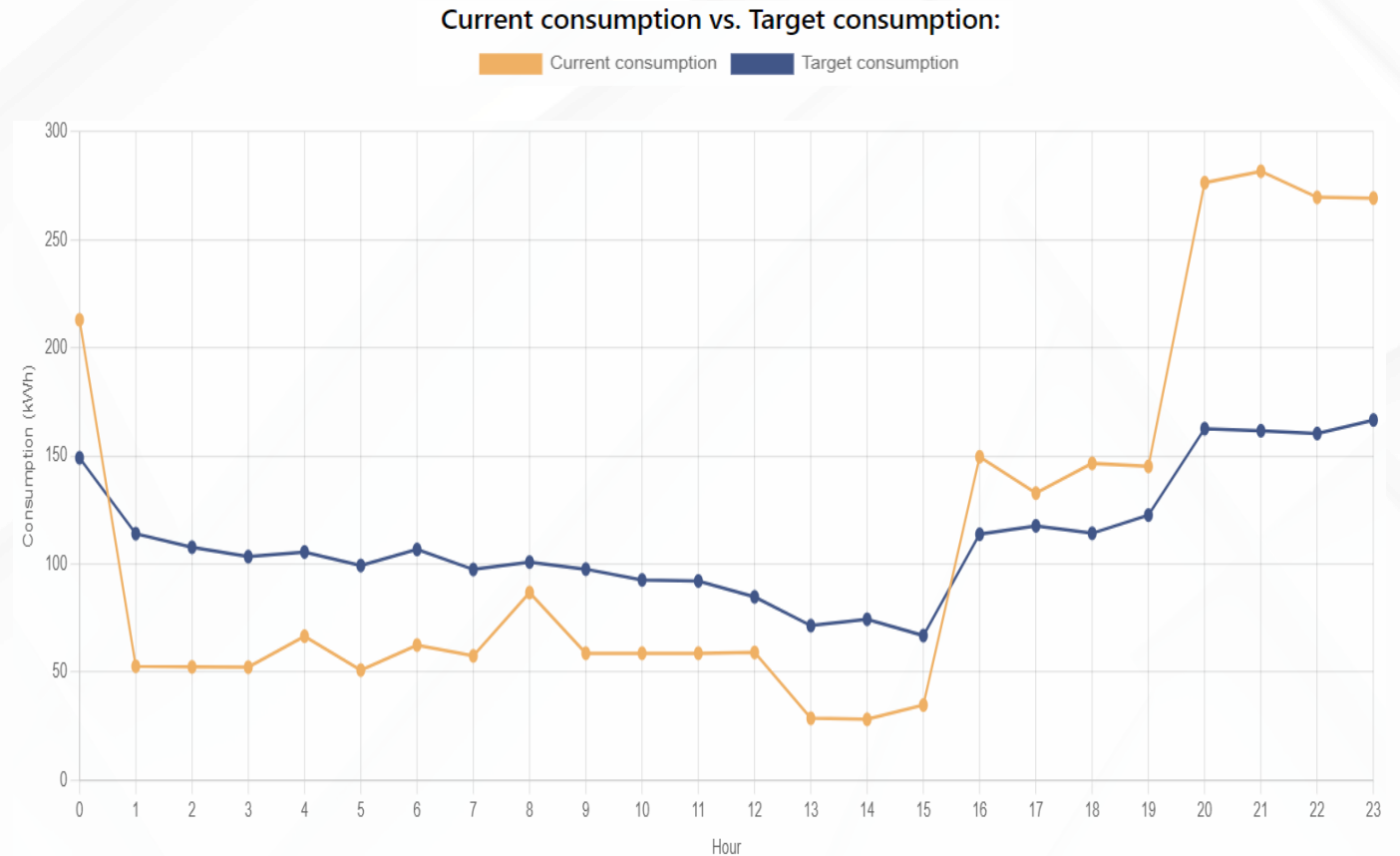


Experimental setup

Dataset definition

1. Survey dataset

- **39 respondents**, students
- focused on usage patterns in **dormitories**
- target curve: **flattened**
- devices included:
 - *heating**
 - *cooling**
 - *laundry*
 - *water-heating*
 - *cooking*
 - *lighting*
 - *fridge**
 - *microwave*
- no. devices: **312**
- dimension of the search space: **7488**



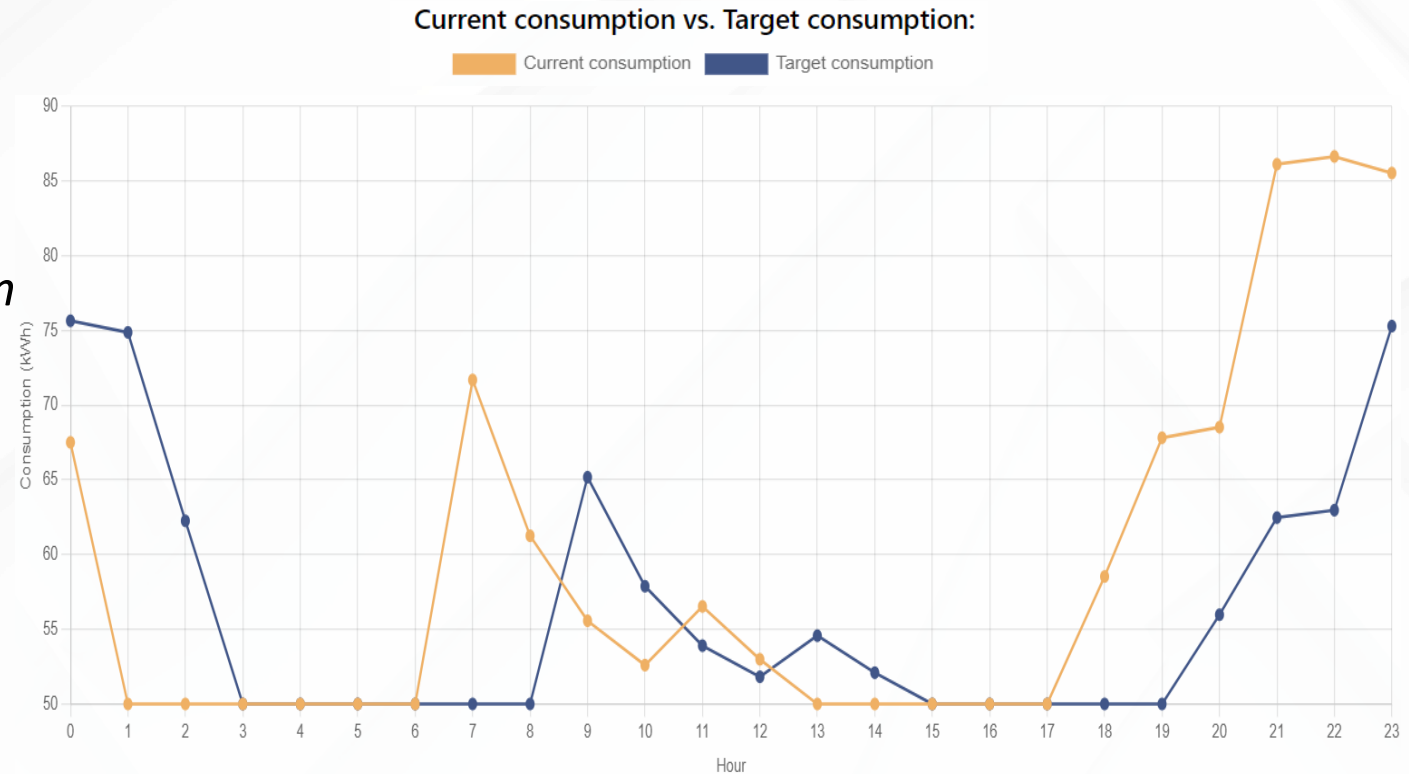
* non-deferrable

Experimental setup

Dataset definition

2. Generated dataset

- **proprietary script** generated datasets
- 2 and 4 number of tenants
- 2 consumption patterns considered:
 - intervals: *8am - 12pm, 9pm - 12am*
 - intervals: *7am- 8am, 6pm - 11pm*
- devices included:
 - *fridge**
 - *lighting*
 - *laptop*
 - *microwave*
- **progressively increasing** no. apartments: 50 → 400, step 50



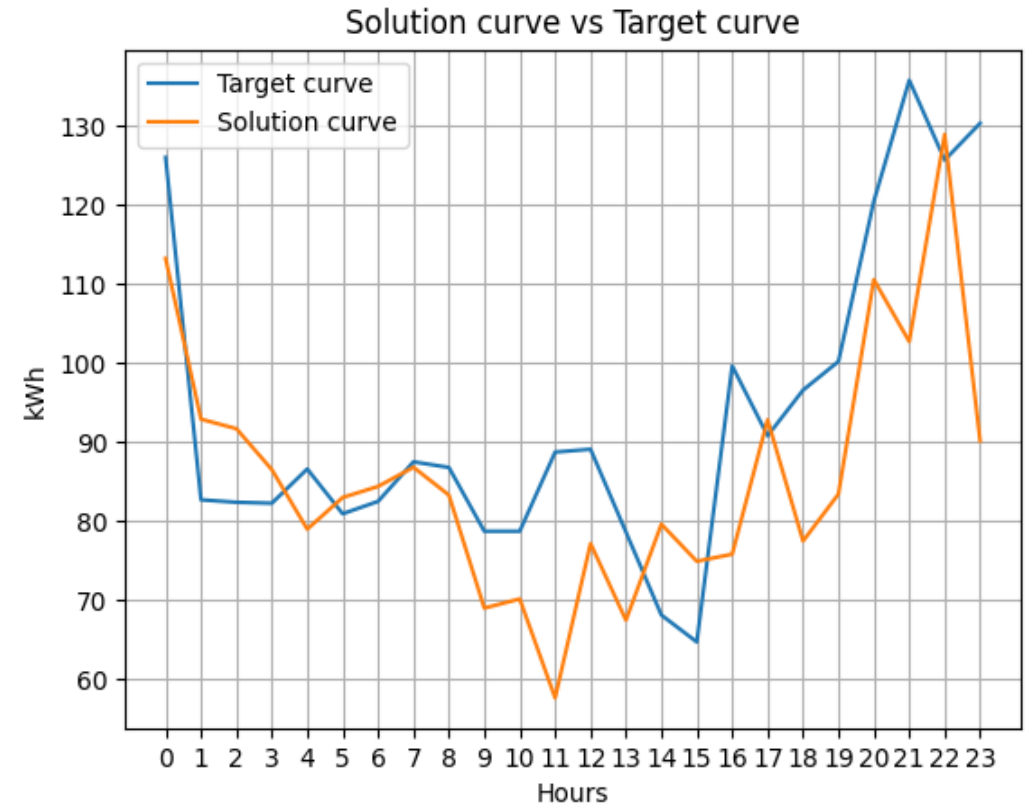
* non-deferrable

Experimental results

Objective 1 results – after hyperparameters tuning

- magnitude is approximated, the **shape not**
- **hypotheses:**
 1. the objective function is flawed
 2. with the current dataset the target can not be achieved

parameter name	value
number of epochs	20
population	25
comfort trade-off	0.35
comfort penalty base	2
discrete transform threshold	0.5

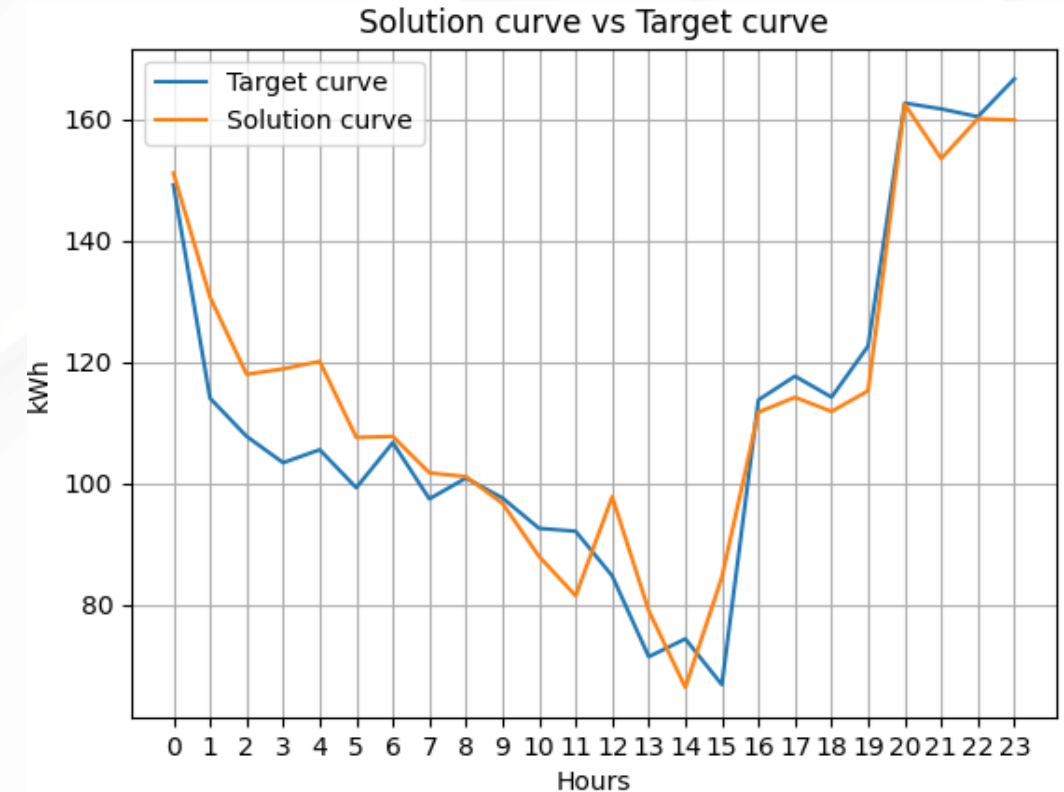


result: distance: 266 kWh

Experimental results

Objective 1 results – Euclidean distance vs. Pearson correlation coefficient

- **only the approximation** objective is considered
- objective function is the result of a previous run
→ *rule out that the curve cannot be approximated with the given dataset*
- evaluation functions tested:
 - only Euclidean distance
 - only the Pearson correlation coefficient
 - mixing Euclidean distance ($NF_{\text{Euclidean}}$) and Pearson correlation coefficient (NF_{Pearson}) → **best generalization**
- observation: the **Euclidean distance** reduced indeed the **offsets**, but the **shape** is best approximated by the **Pearson correlation coefficient**



parameter: $w_{\text{Euclidean}} = 0.25$
 $w_{\text{Pearson}} = 0.75$
result: **distance: 160 kWh**

Experimental results

Objective 1 results – HHO → improved HHO

- all objectives considered
- hypothesis: *too many optimization parameters (7488)*
- **solution:** reduce the number of parameters by running a HHO optimization for each hour (opt. params. **312** at once)
- leverage **parallelism**: execute in parallel the optimizations
 - from **49 seconds** to **1.27 seconds** running time

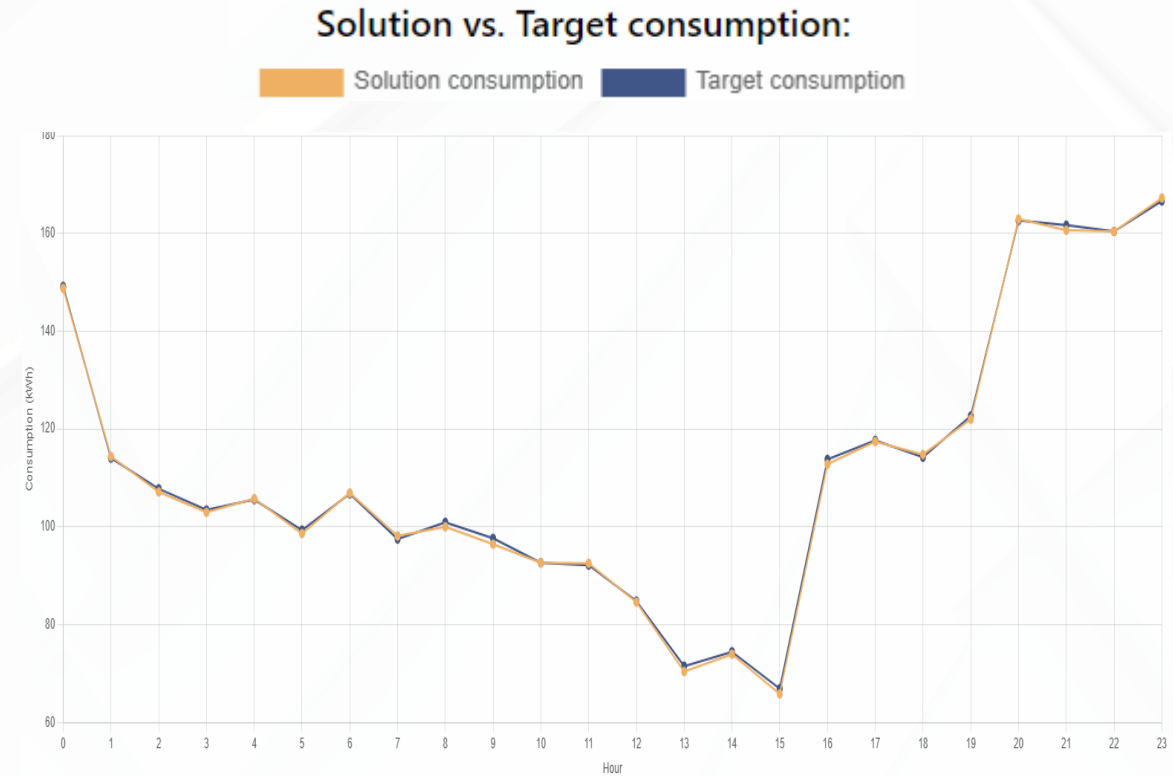


Figure 8. Improved HHO approximation result
comfort trade-off: 0
→ distance: **13.36 kWh**

Experimental results

Greedy algorithm

- I developed a greedy algorithm for benchmark
 - based on the knapsack problem
- it produces excellent results, but is **unable to prioritize objectives**
- running time: **0.034 seconds**

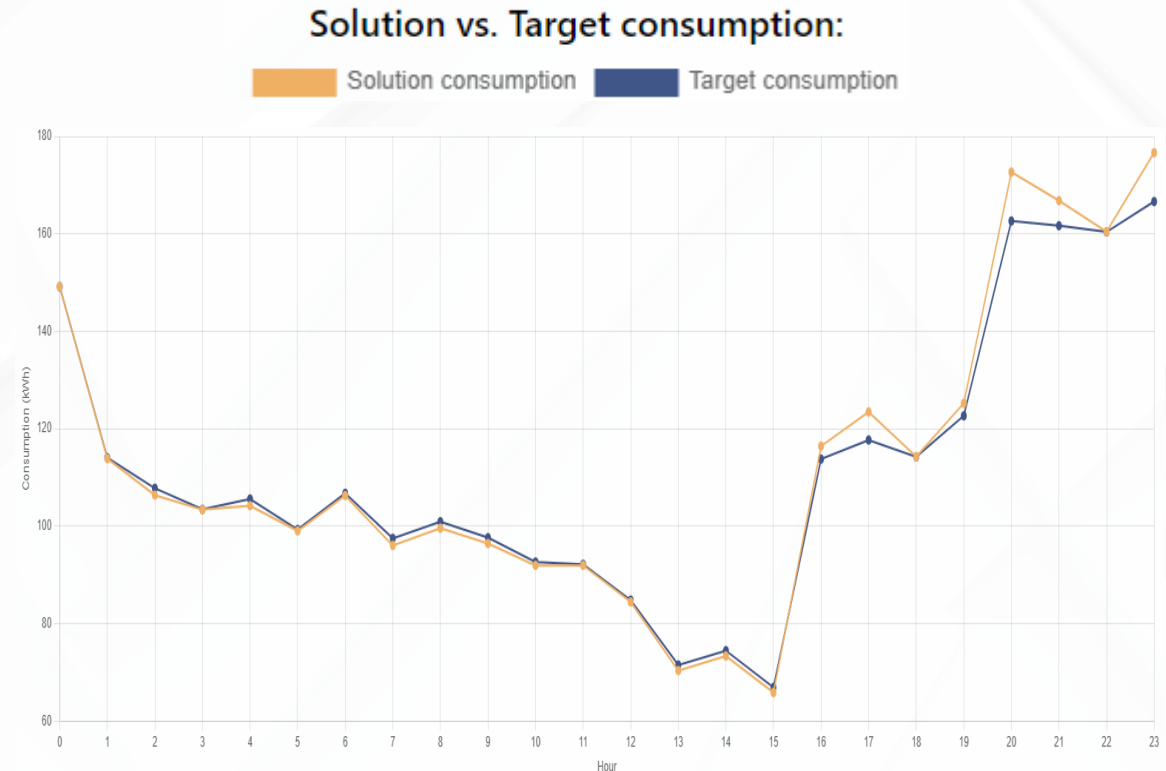


Figure 9. Greedy algorithm approximation result

→ distance: **48.72 kWh**

→ discomfort coeff.: **4194**

Experimental setup

Objective 2 results – Genetic algorithm (GA) + Harris Hawks Optimization (HHO)

- subset is selected by the GA → devices from selected apartments are scheduled by HHO
- as expected: increased running time
- approximation objective is fulfilled to the same extent
- comfort objective is greatly improved

Running time: 00:04:08	
No. apartments selected:	19
Apartments included	[0, 1, 2, 4, 5, 6, 8, 10, 11, 15, 16, 17, 18, 19, 20, 21, 22, 24, 26]
Ap. selected from total	48.7%
Distance (kWh)	24.85
Discomfort metric	1371

Table 2. Numerical results of the Bi-Level optimization strategy

Experimental results

Results comparison - basic HHO vs. improved HHO vs. greedy algorithm vs. bi-level optimization

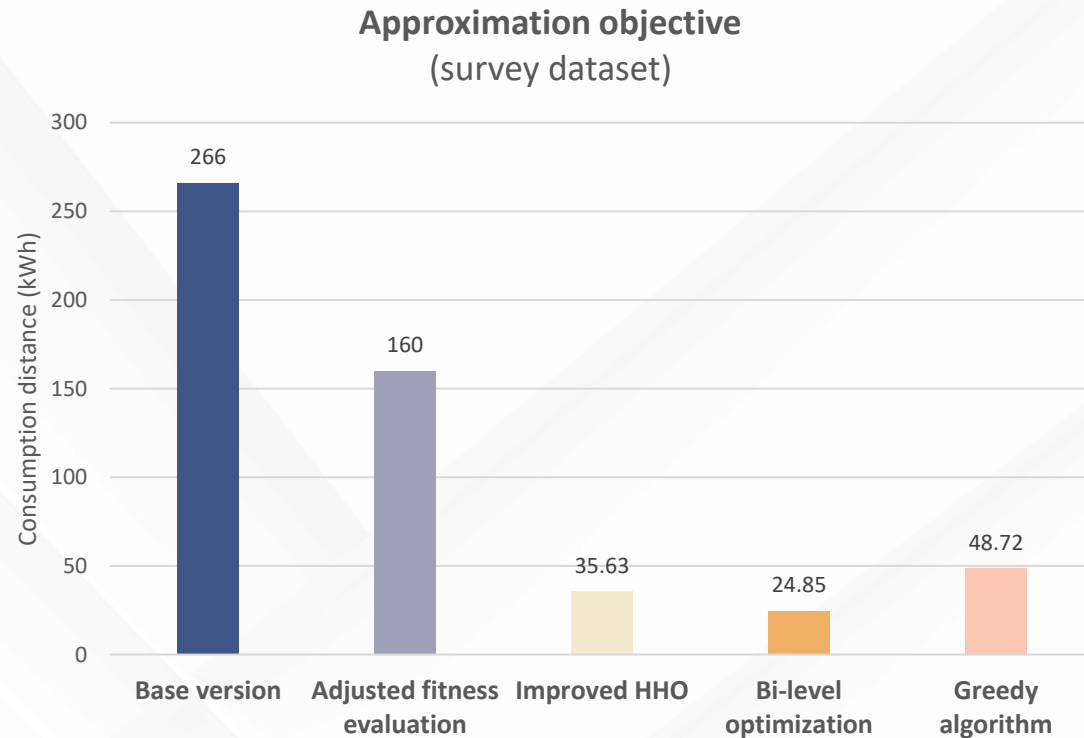


Figure 11. Approximation fitness comparison on the survey dataset

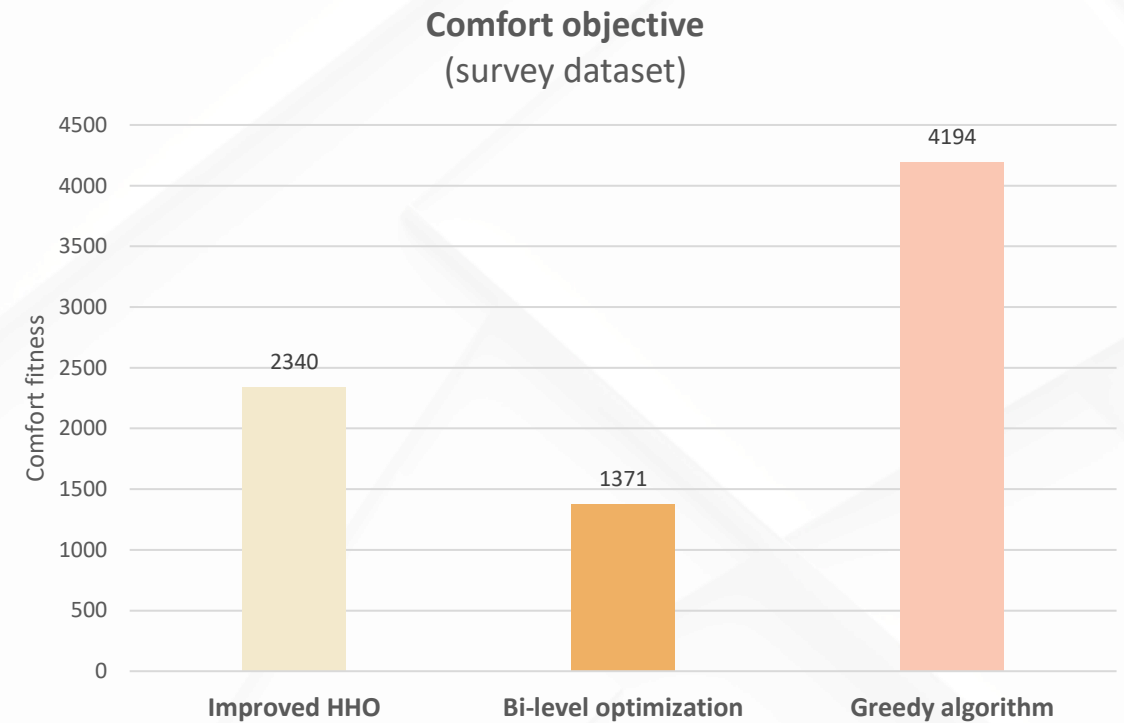


Figure 12. Comfort fitness comparison on the survey dataset

Experimental results

Results comparison - basic HHO vs. improved HHO vs. greedy algorithm vs. bi-level optimization

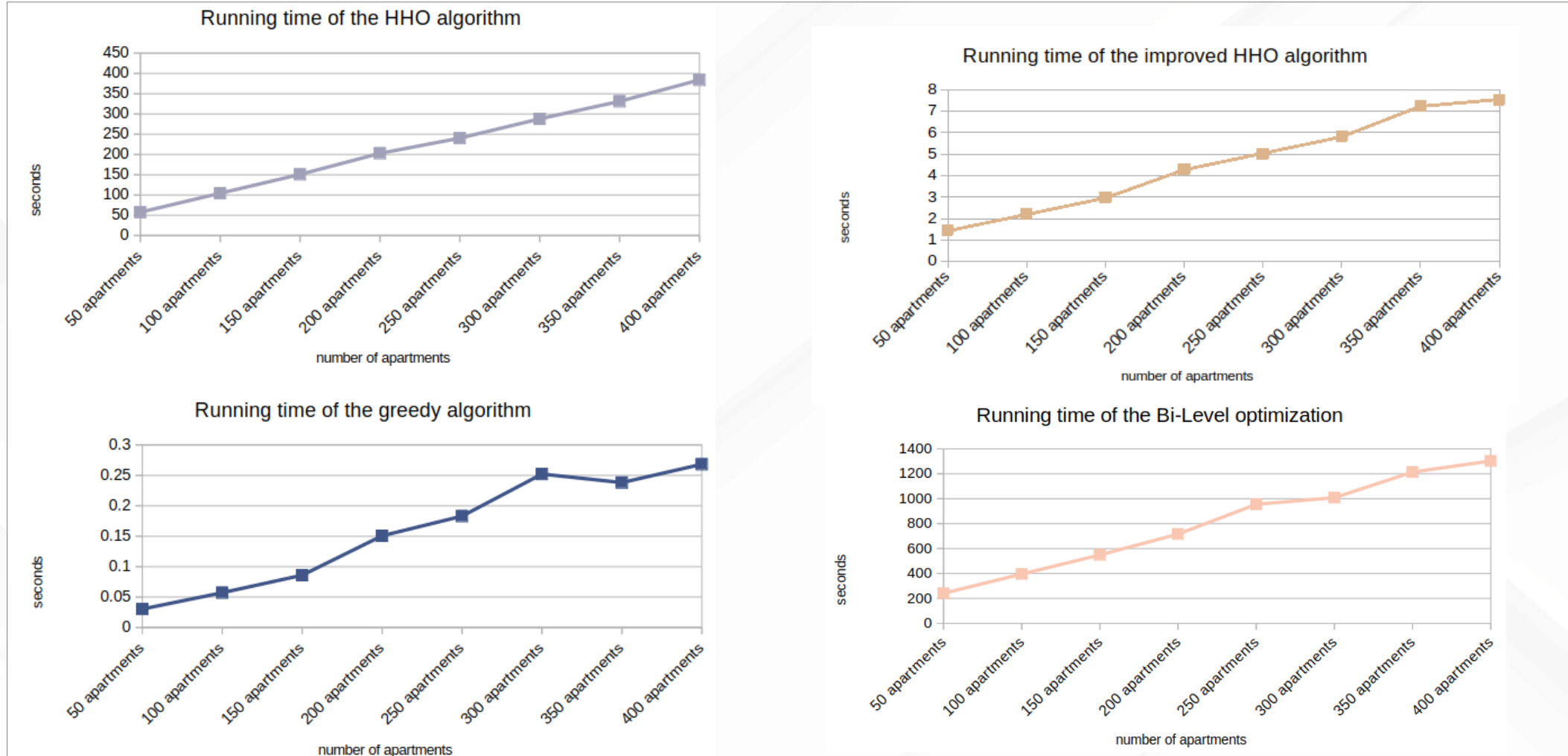


Figure 10. Running time comparison between the HHO, improved HHO, and greedy algorithms

Experimental results

Results comparison - basic HHO vs. improved HHO vs. greedy algorithm vs. bi-level optimization

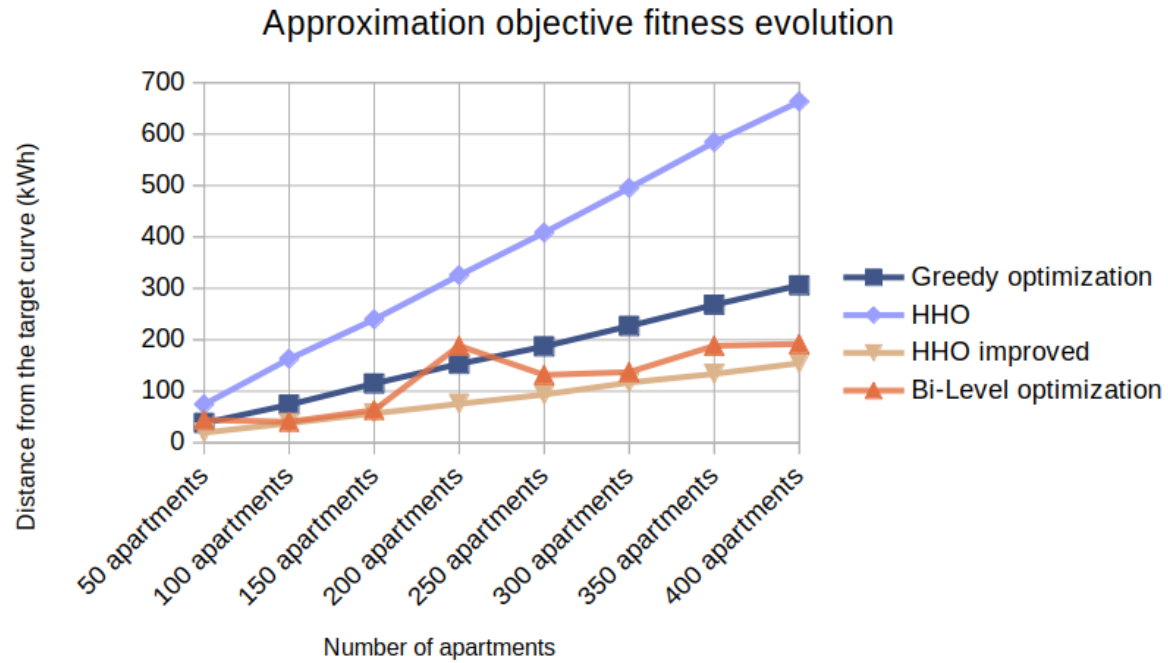


Figure 11. Approximation objective comparison between the 4 implementation variants

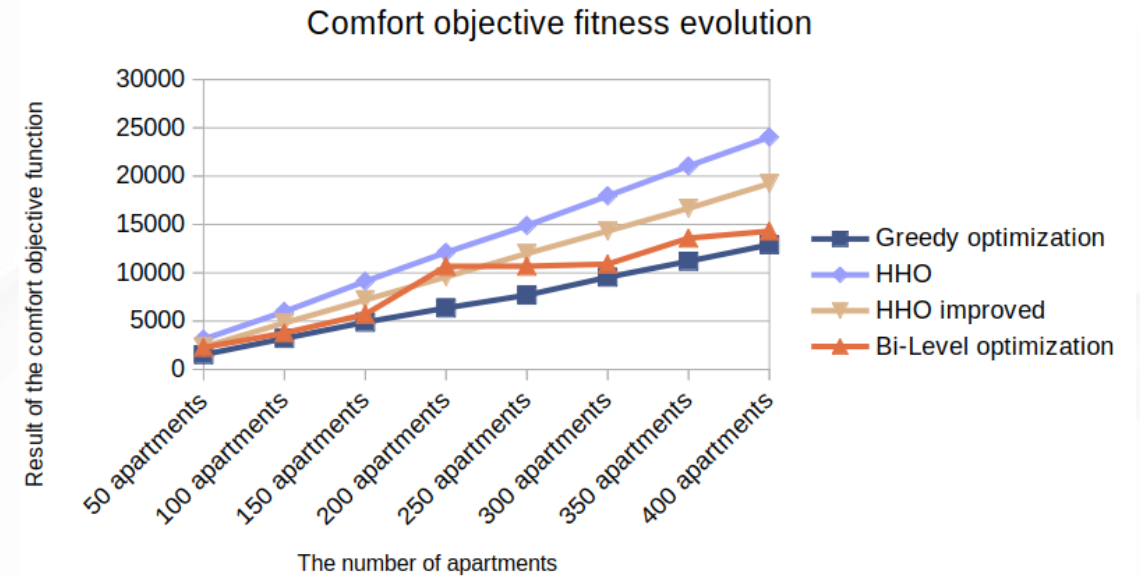


Figure 12. Comfort objective comparison between the 4 implementation variants

Conclusion

Main takeaways

- with a **reduced number of parameters** and **parallelization** the performance of the HHO algorithm was **significantly improved**
- the HHO algorithm had a **greater running time** compared to the greedy algorithm
- the HHO **balanced the objectives better**, than the greedy algorithm
- the HHO was **more consistent** with different target curves, than the greedy algorithm

Main takeaways cont'd

- choosing a subset **significantly reduced the impact on comfort** while having similar results
- HHO is more **suitable for multi-objective problems**
- greedy **performs better for single-objective problems**
- the **increased running time** of the Bi-Level strategy is **acceptable compared** to the **resulting benefits**

Conclusion

Personal contributions

1. **Review** and analysis of **state-of-the-art solutions**
2. **Designing a bi-level optimization strategy** to solve the multi-objective problem
3. **Defining the scheduling problem** & implementing it using **Harris Hawks Optimization algorithm**
4. Identifying and **correcting the shortcomings** of HHO
5. **Hyper-parameter tuning**
6. Developing a **greedy algorithm** for benchmark
7. Developing a **visual interface** for accessing functionalities

Further developments

1. Introduce **more classes** for categorizing **devices** → **more constraints**
2. Have **distinct comfort trade-off** for each apartment
3. Analyze the **cause of poor performance** with many optimization parameters
4. Extend the objectives to include **cost reduction**
5. Enhance the user experience by letting tenants introduce their preferences for the optimization

The work presented in this thesis is **part of a scientific article** that is in progress and will be submitted to a journal

Bibliography

1. A. Shewale, A. Mokhade, N. Funde, and N. D. Bokde, "An overview of demand response in smart grid and optimization techniques for efficient residential appliance scheduling problem," *Energies*, 2020Z. Luo, J. Peng, and R. Yin, "Many-objective day-ahead optimal scheduling of residential flexible loads integrated with stochastic occupant behavior models," *Applied Energy*, 2023
2. Y. Wei, Q. Meng, F. Zhao, I. Yu, L. Zhang, and I. Jiang, "Direct load control-based optimal scheduling strategy for demand response of air-conditioning systems in rural building complex," *Building and Environment*, 2024
3. D. H. Muhsen, H. T. Haider, Y. Al-Nidawi, and G. G. Shaya, "Operational scheduling of household appliances by using triple-objective optimization algorithm integrated with multi-criteria decision making," *Sustainability*, 2023
4. L. Guanghua, S. Ullah, G. Hafeez, and H. Alghamdi, "An application of heuristic optimization algorithm for demand response in smart grids with renewable energy," *AIMS Mathematics*, 2024
5. A. A. Dashtaki, M. Khaki, M. Zand, M. A. Nasab, P. Sanjeevikumar, T. Samavat, M. A. Nasab, and B. Khan, "A day ahead electrical appliance planning of residential units in a smart home network using its-bf algorithm," *International Transactions on Electrical Energy Systems*, 2022
6. Z. Chen, Y. Chen, R. He, J. Liu, M. Gao, and L. Zhang, "Multi-objective residential load scheduling approach for demand response in smart grid," *Sustainable Cities and Society*, 2021

Bibliography

7. Y. Liu, H. Li, J. Zhu, Y. Lin, and W. Lei, “Multi-objective optimal scheduling of household appliances for demand side management using a hybrid heuristic algorithm,” Energy, 2022
8. Background images are the courtesy of www.freepik.com

**Thank you for your
attention!**



Cluj-Napoca, 17.07.2024



Appendix

Objective 1 results – HHO → improved HHO

Solution vs. Target consumption:

■ Solution consumption ■ Target consumption

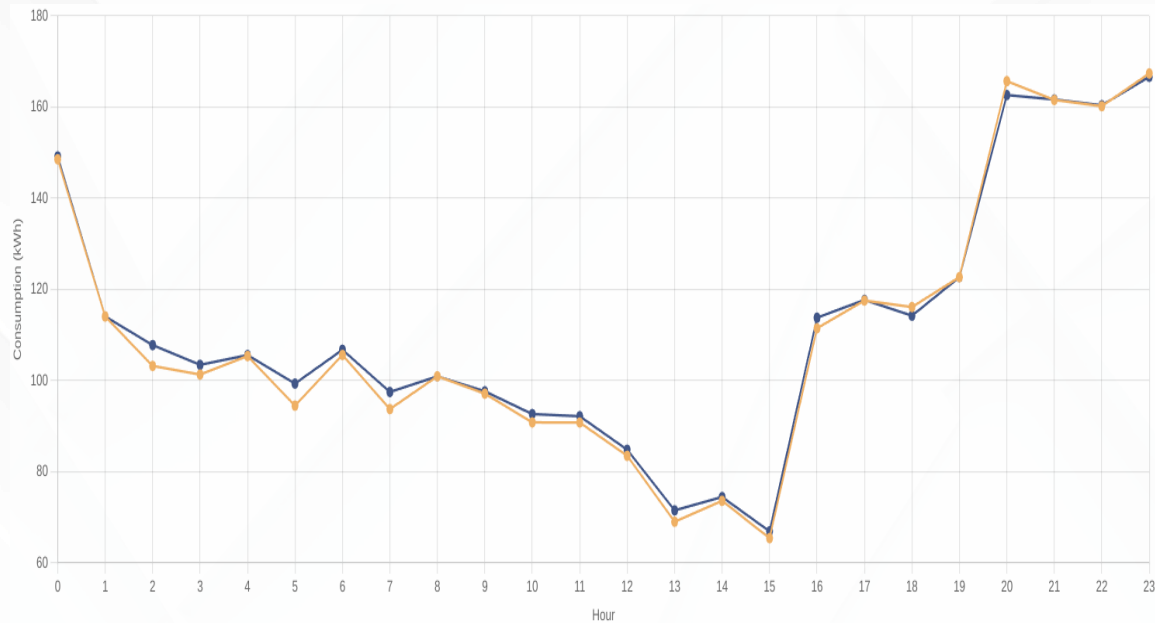
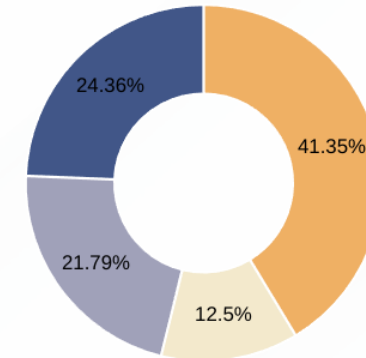


Figure 12. Improved HHO approximation result
comfort trade-off: 0.4
→ distance: **35.63 kWh**
→ discomfort coeff.: **2340**

% of devices having schedule in comfort interval:

■ 100%-75% in comfort interval ■ 75%-50% in comfort interval ■ 50%-25% in comfort interval ■ 25%-0% in comfort interval



% of devices having comfort interval utilized:

■ 100%-75% in time interval ■ 75%-50% in time interval ■ 50%-25% in time interval ■ 25%-0% in time interval

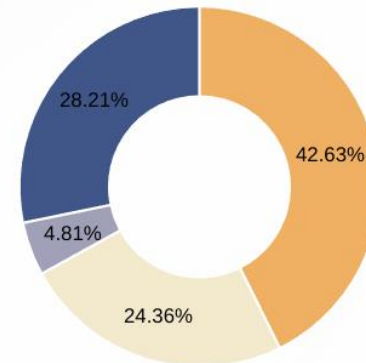


Figure 13. Improved HHO comfort result

Appendix

Objective 1 results – improved HHO trade-off capabilities

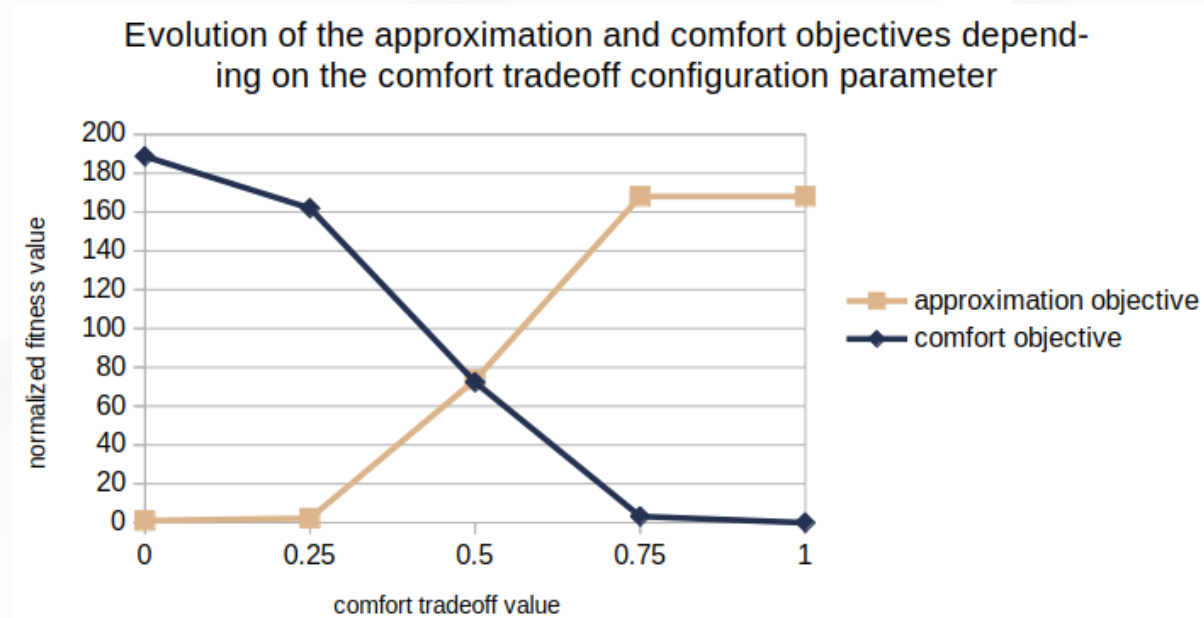
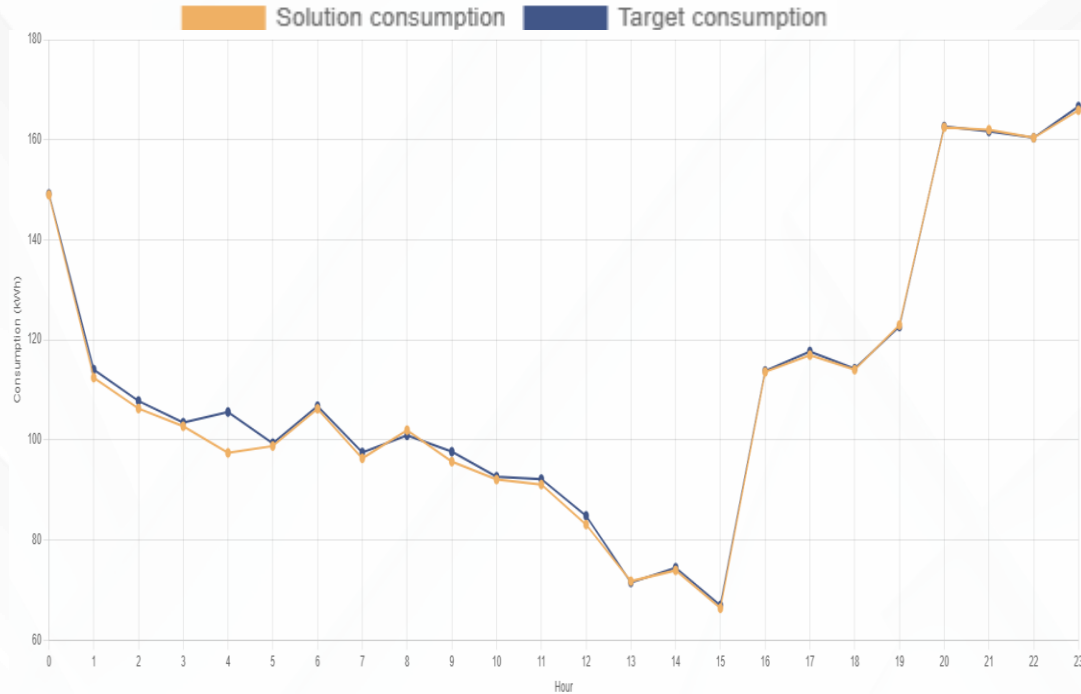


Figure 14. Improved HHO's ability to balance the objectives

Appendix

Objective 2 results – Genetic algorithm (GA) + Harris Hawks Optimization (HHO)

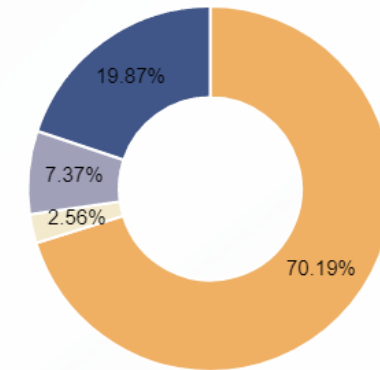
Solution vs. Target consumption:



→ distance: **24.85 kWh**
→ discomfort coeff.: **1371**

% of devices having schedule in comfort interval:

100%-75% in comfort interval 75%-50% in comfort interval 50%-25% in comfort interval 25%-0% in comfort interval



% of devices having comfort interval utilized:

100%-75% in time interval 75%-50% in time interval 50%-25% in time interval 25%-0% in time interval

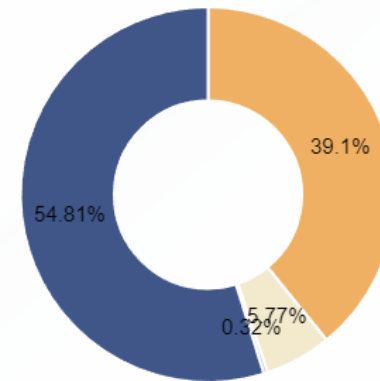
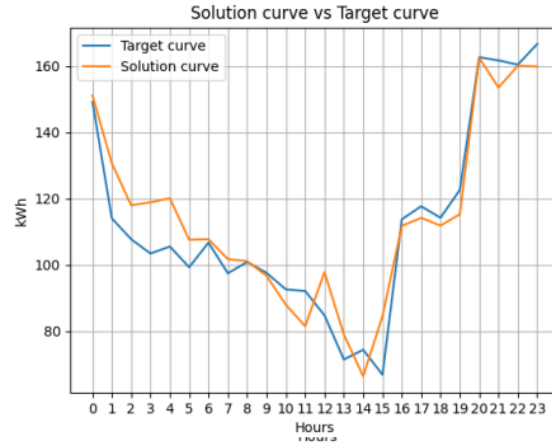


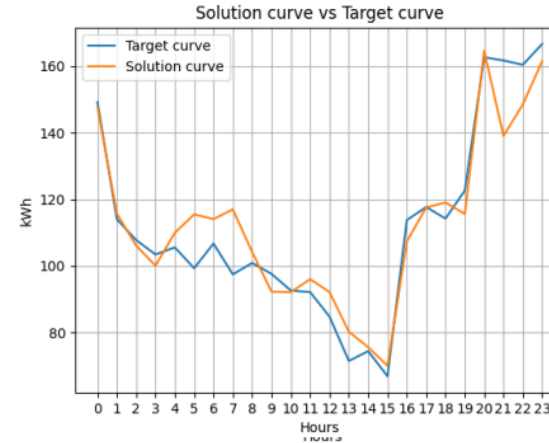
Figure 19. Results of the Bi-Level optimization strategy

Appendix

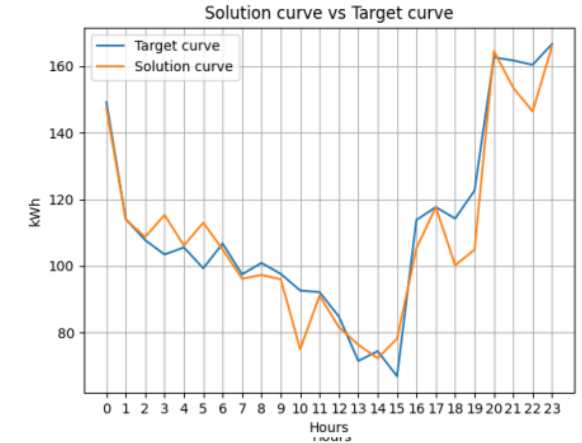
Objective 1 – Fitness evaluation function analysis



(a) Euc. dist. weight: 0.25
Pearson corr. weight: 0.75
Pearson correlation: 0.95
Distance: 160 kWh



(b) Euc. dist. weight: 0.5
Pearson corr. weight: 0.5
Pearson correlation: 0.949
Distance: 165 kWh



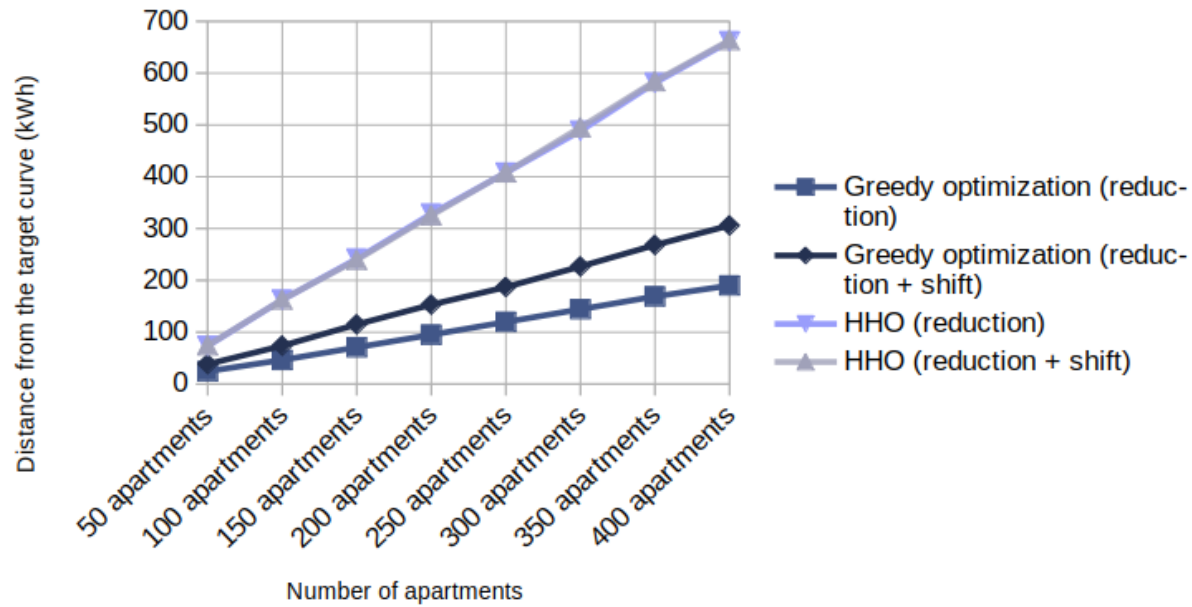
(c) Euc. dist. weight: 0.75
Pearson corr. weight: 0.25
Pearson correlation: 0.95
Distance: 163 kWh

Evaluation metric	The offset given from the target curve obtained using the evaluation metric		Variation ratio
	Euclidean distance	Pearson corr. coeff	
Euclidean distance	84.31 kWh	156 kWh	0.134
Pearson corr. coeff.	76.68 kWh	186.06 kWh	0.335
Mixed evaluation metric	87 kWh	162.66 kWh	0.779
<i>Average</i>	<i>82.66 kWh</i>	<i>168.23 kWh</i>	

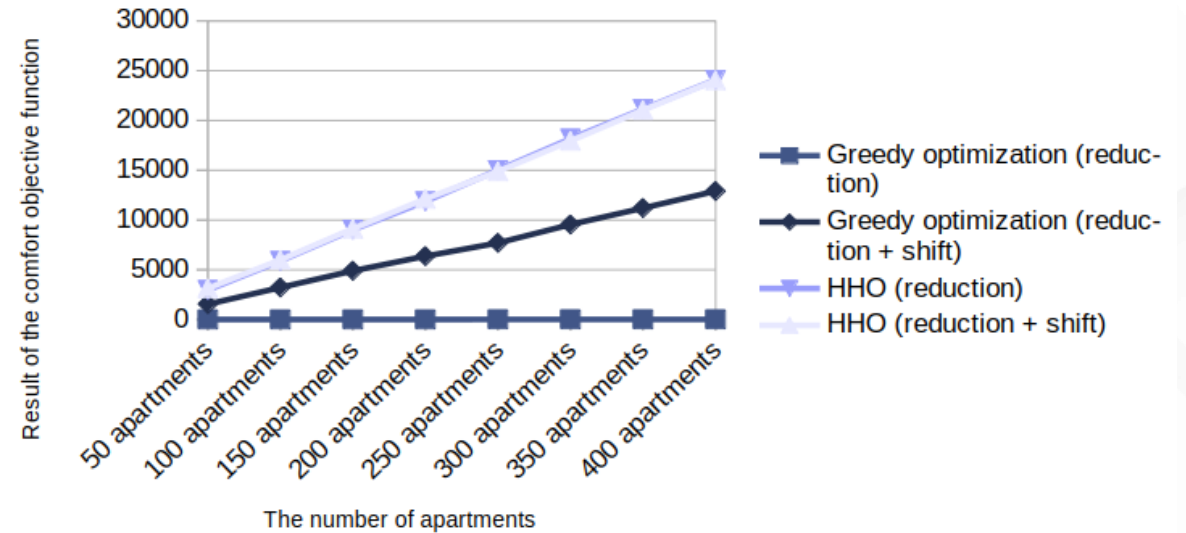
Appendix

Implementation version comparison – 2 types of target curves

Approximation objective fitness evolution

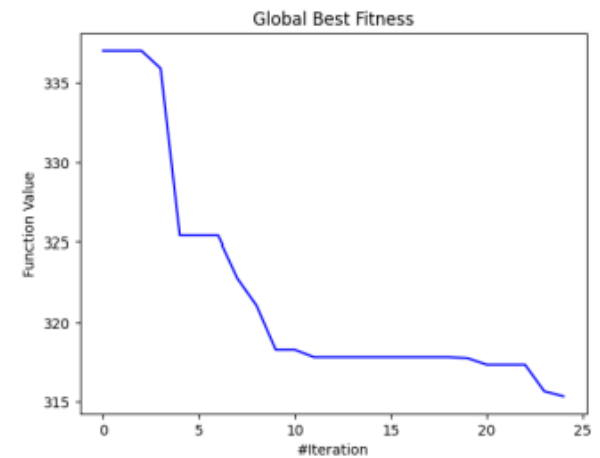
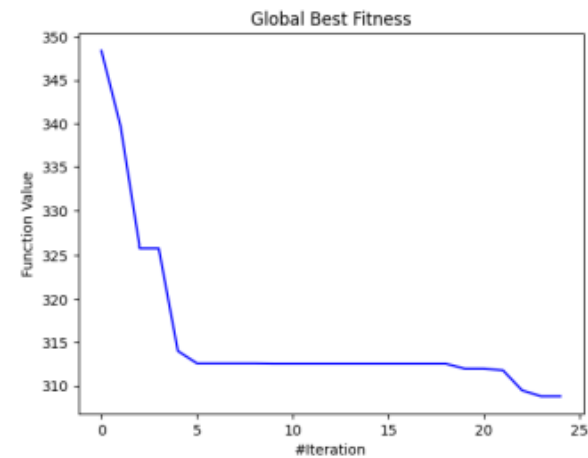
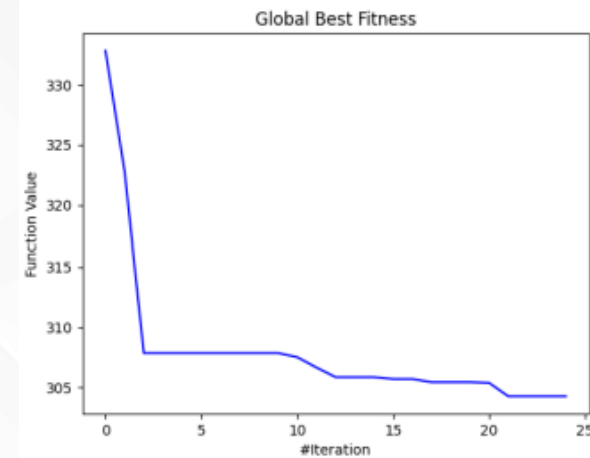
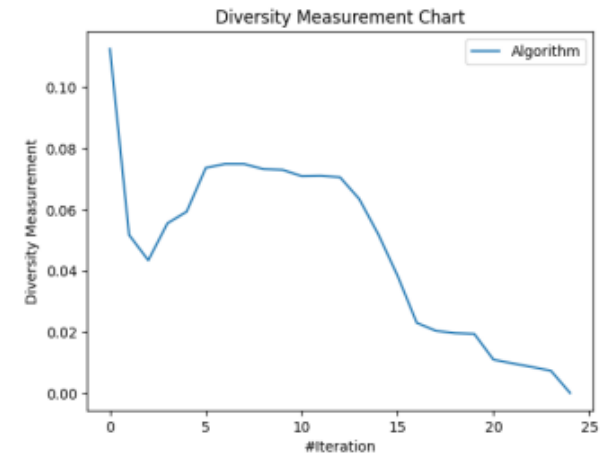
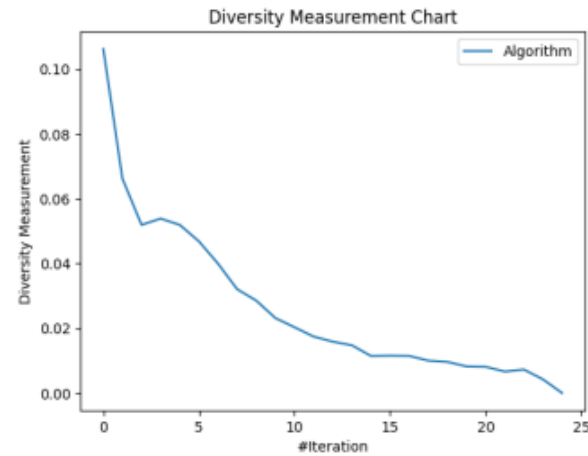
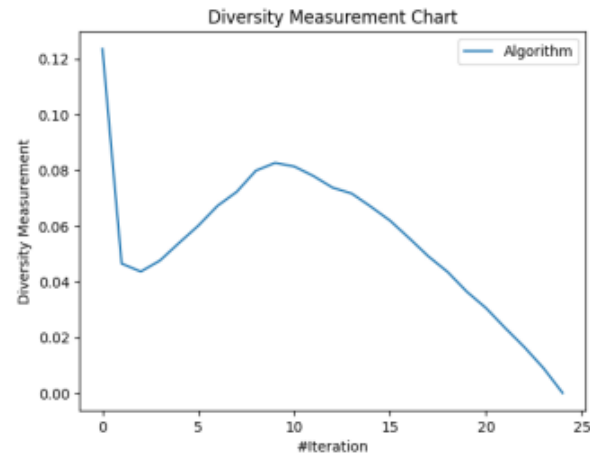


Comfort objective fitness evolution



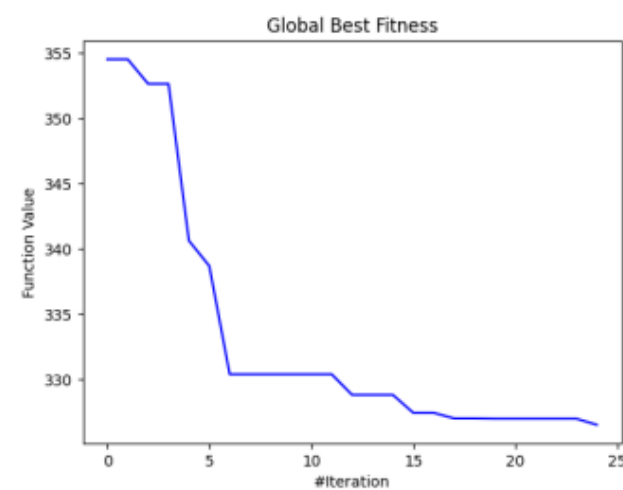
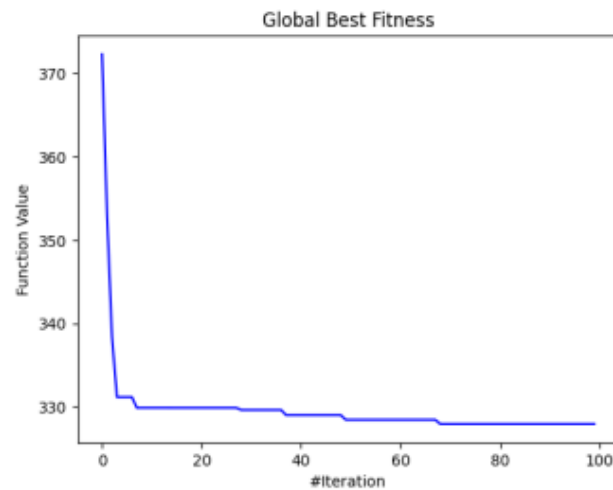
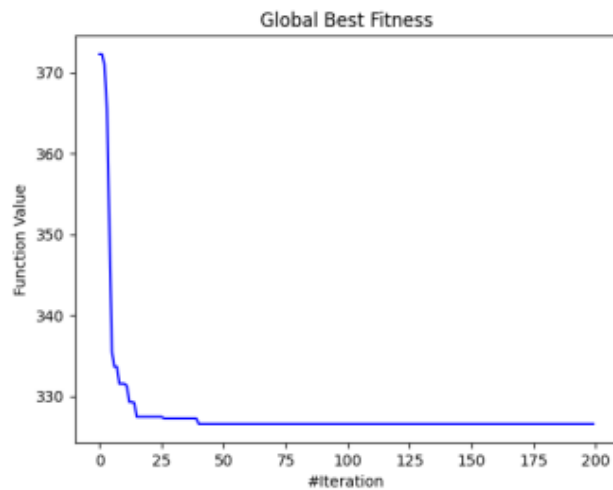
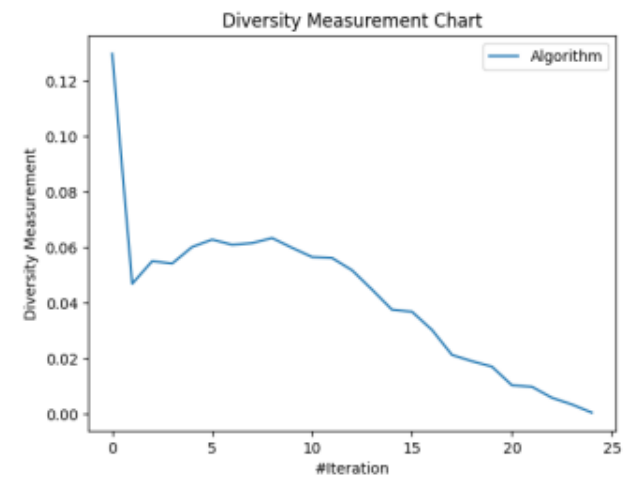
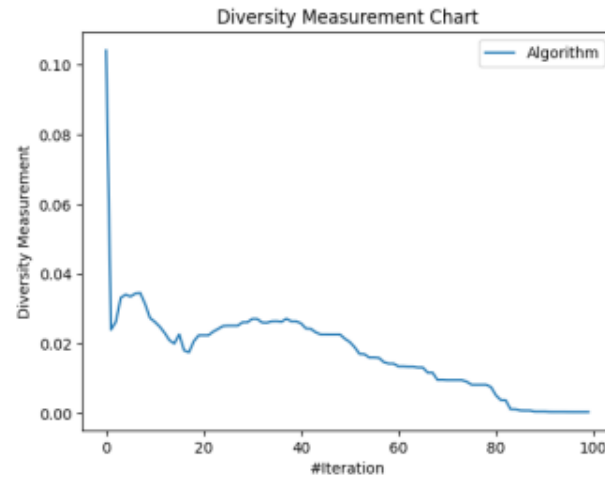
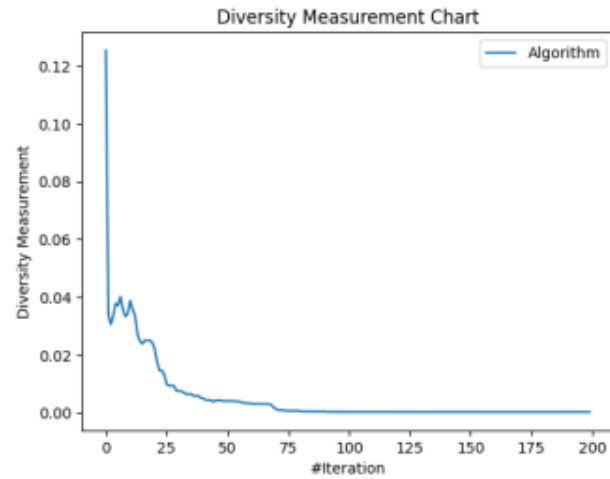
Appendix

HHO - Hyperparameter tuning – population size



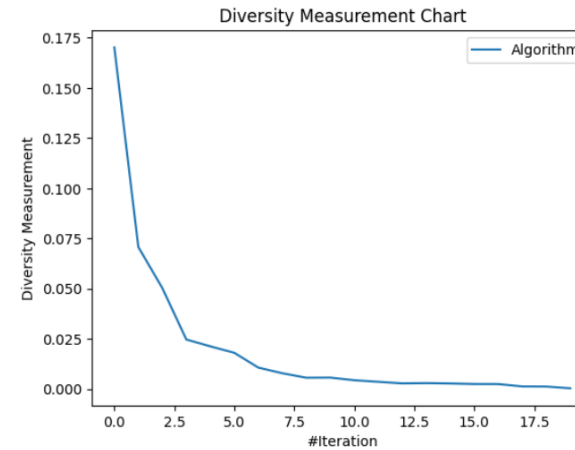
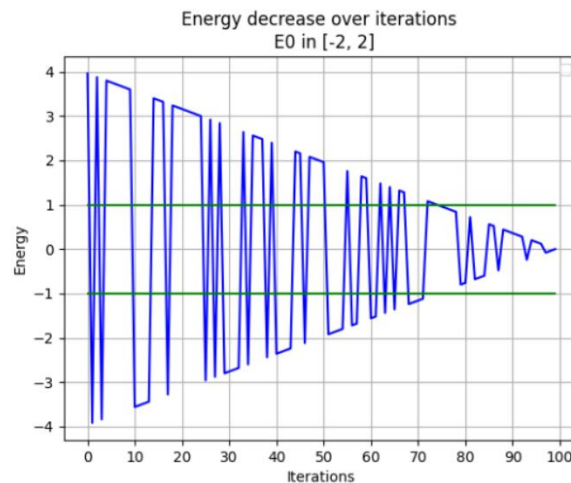
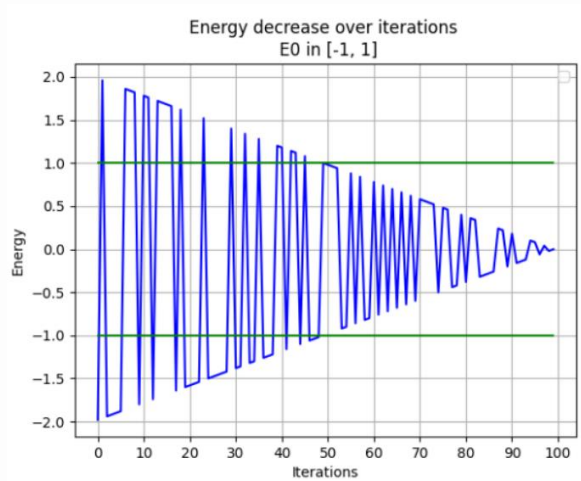
Appendix

HHO - Hyperparameter tuning – epoch number

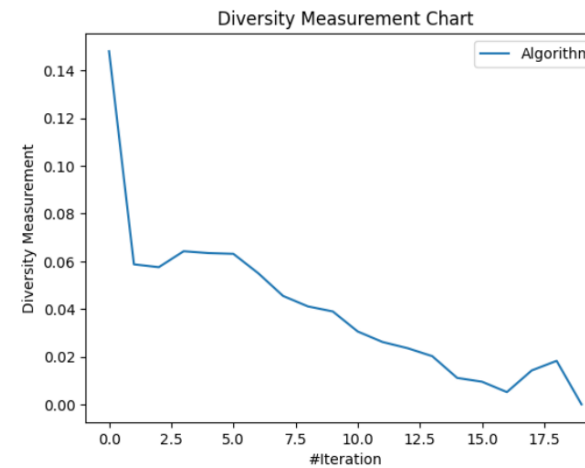


Appendix

HHO - Initial energy adjustment



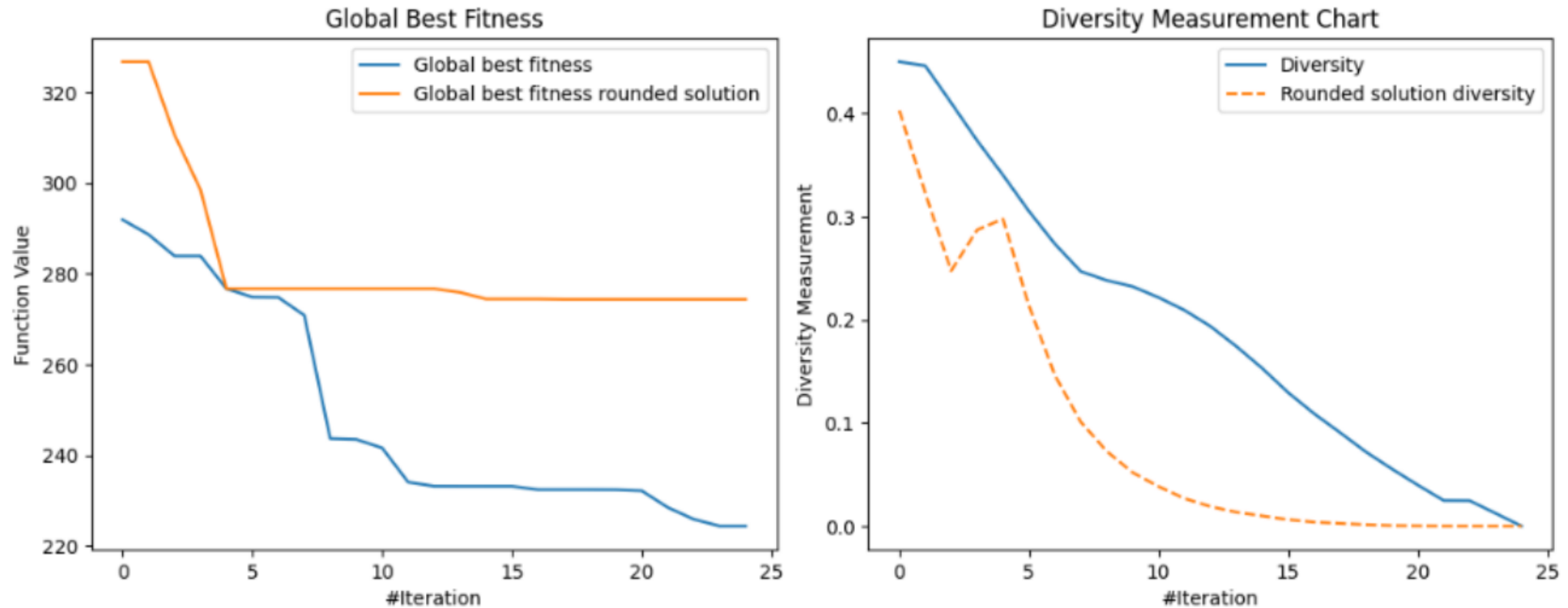
(a) Diversity when E_0 in [-1, 1]



(b) Diversity when E_0 in [-2, 2]

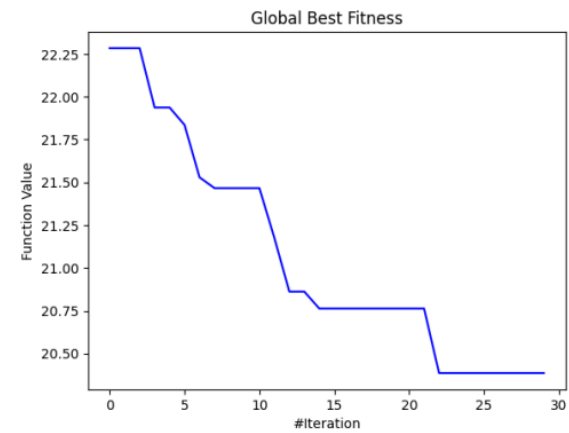
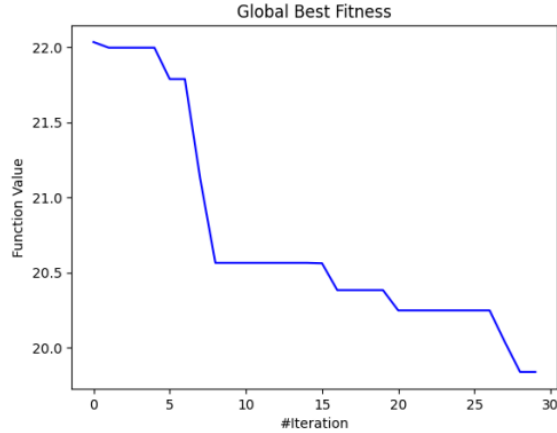
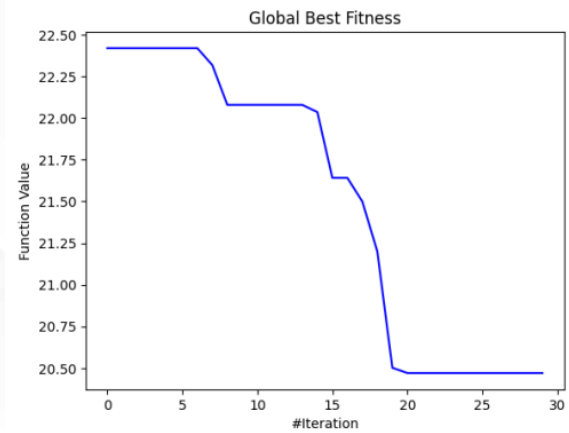
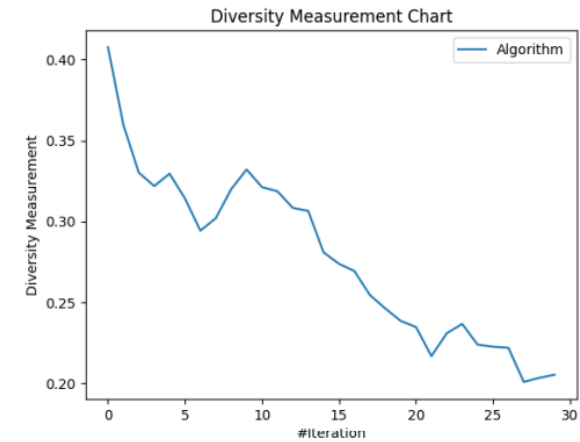
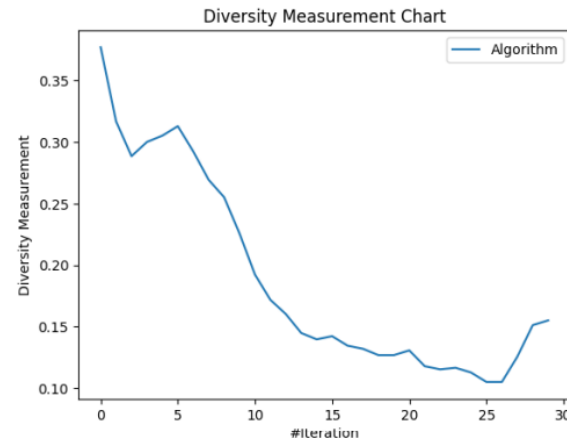
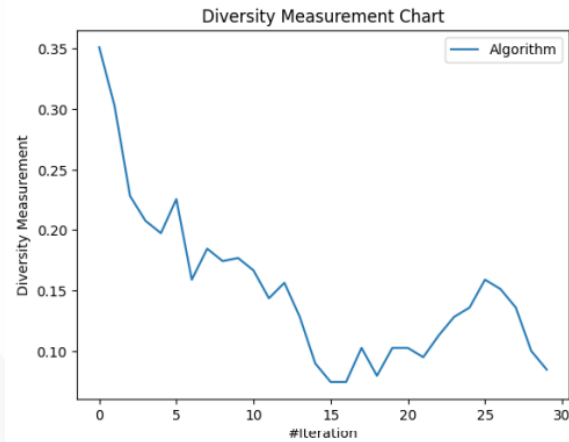
Appendix

HHO - Effect of rounding



Appendix

GA - Hyperparameter tuning – population size



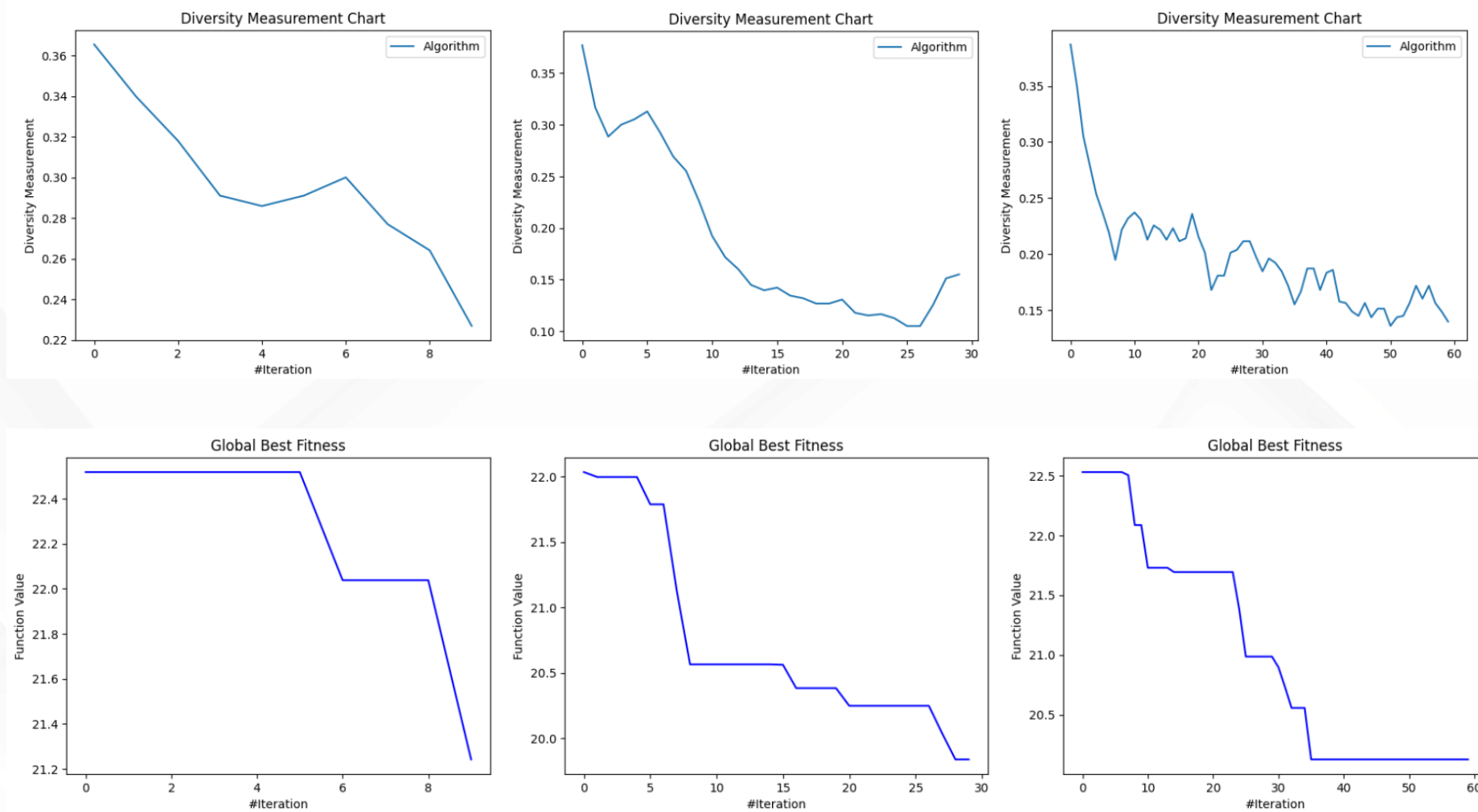
(a) $\text{subs_pop_size} = 10$

(b) $\text{subs_pop_size} = 30$

(c) $\text{subs_pop_size} = 40$

Appendix

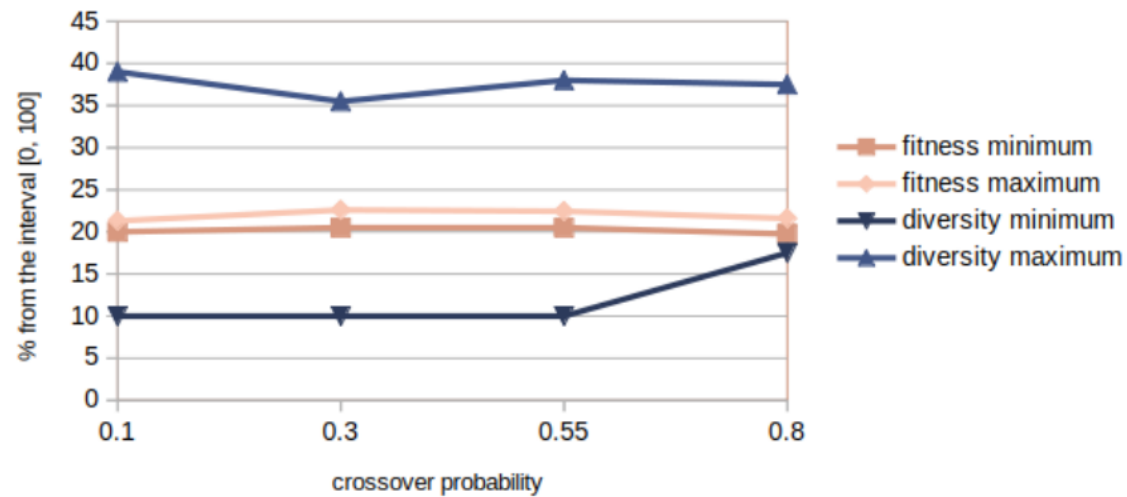
GA - Hyperparameter tuning – epoch number



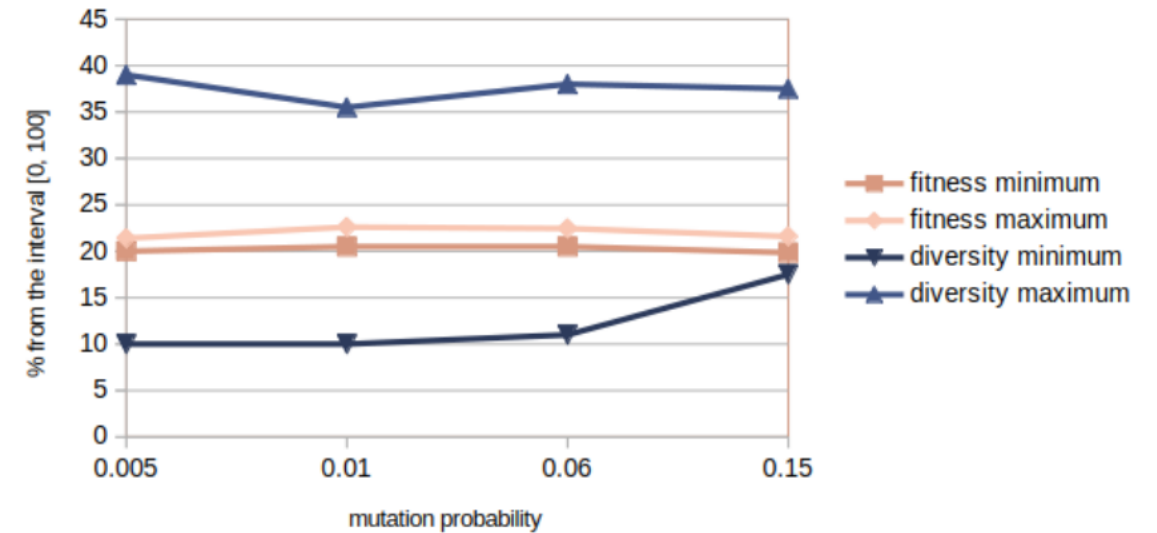
Appendix

GA - Hyperparameter tuning – epoch number

Evolution of fitness and diversity with the crossover probability

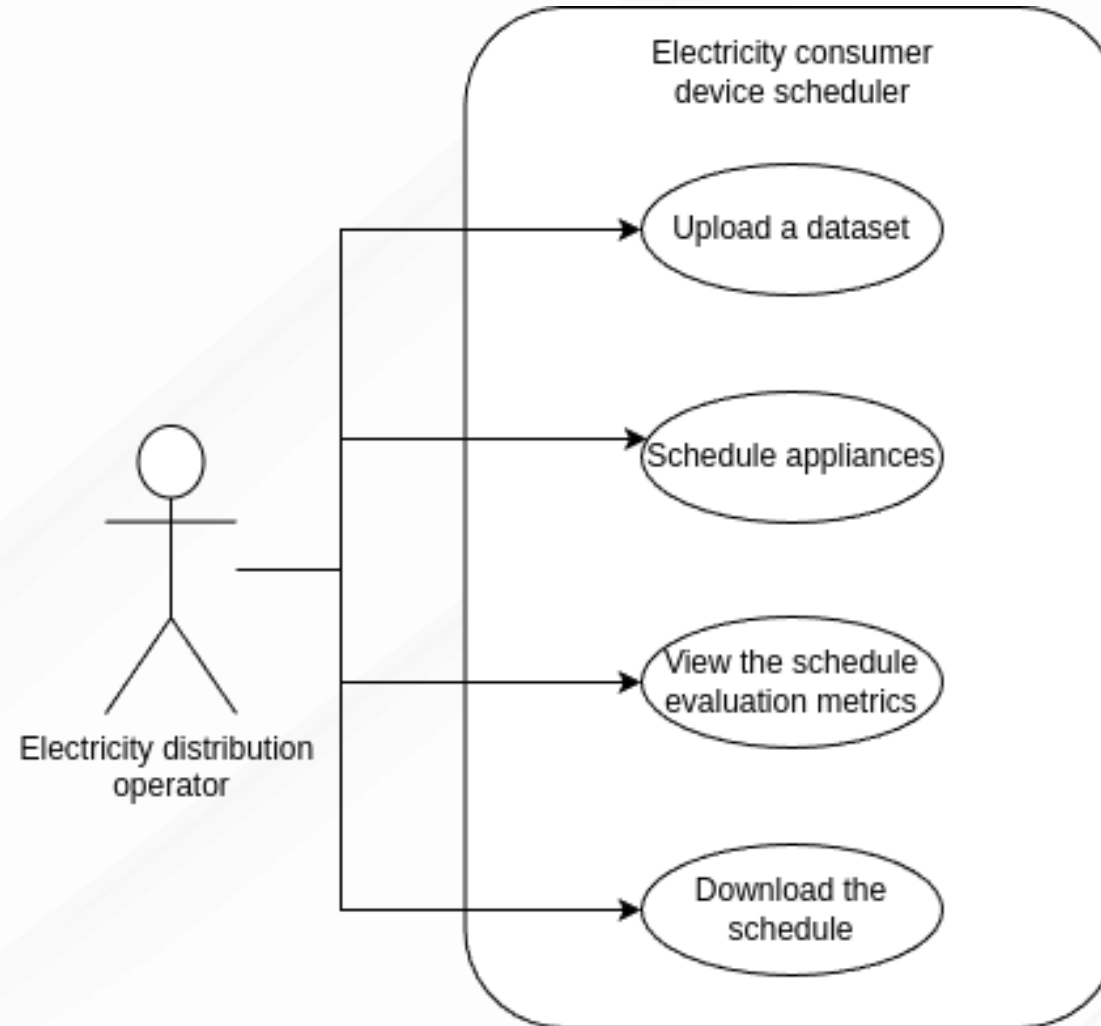


Evolution of fitness and diversity with the mutation probability



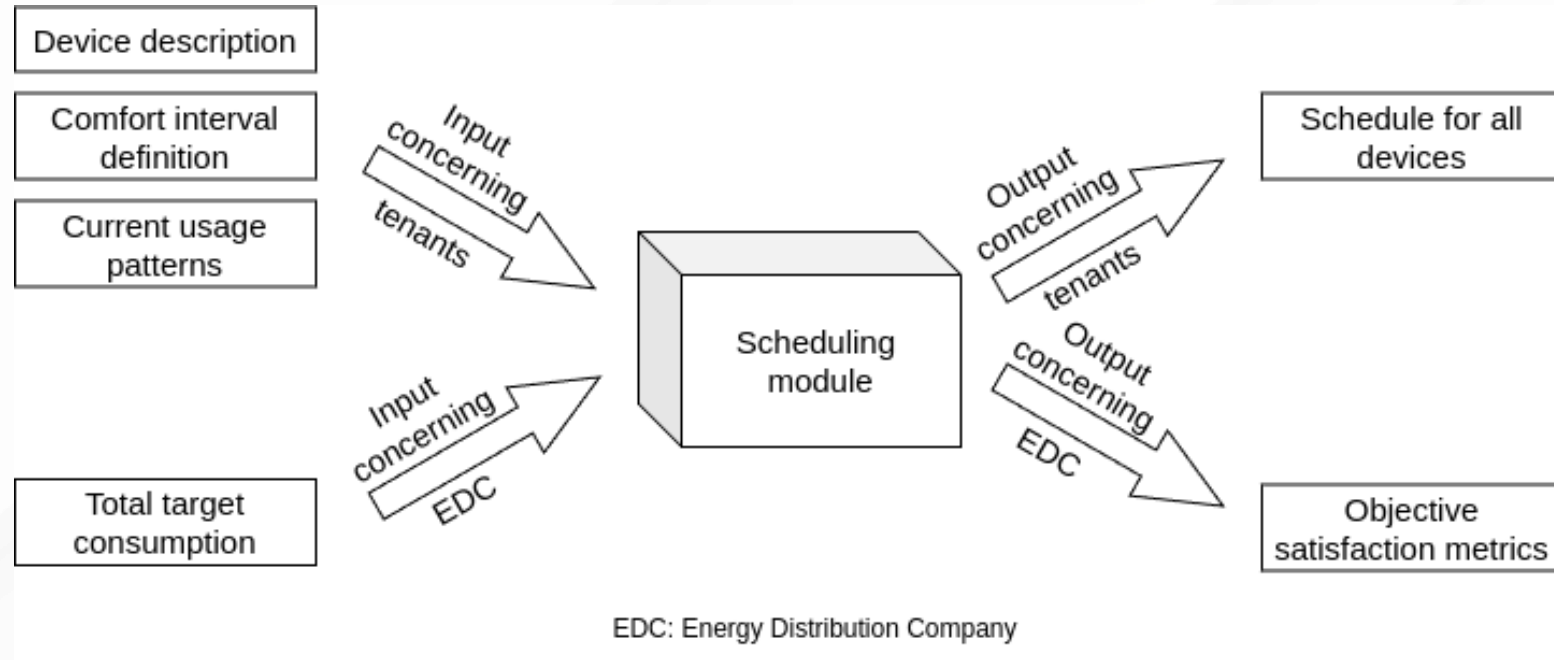
Appendix

Use case diagram



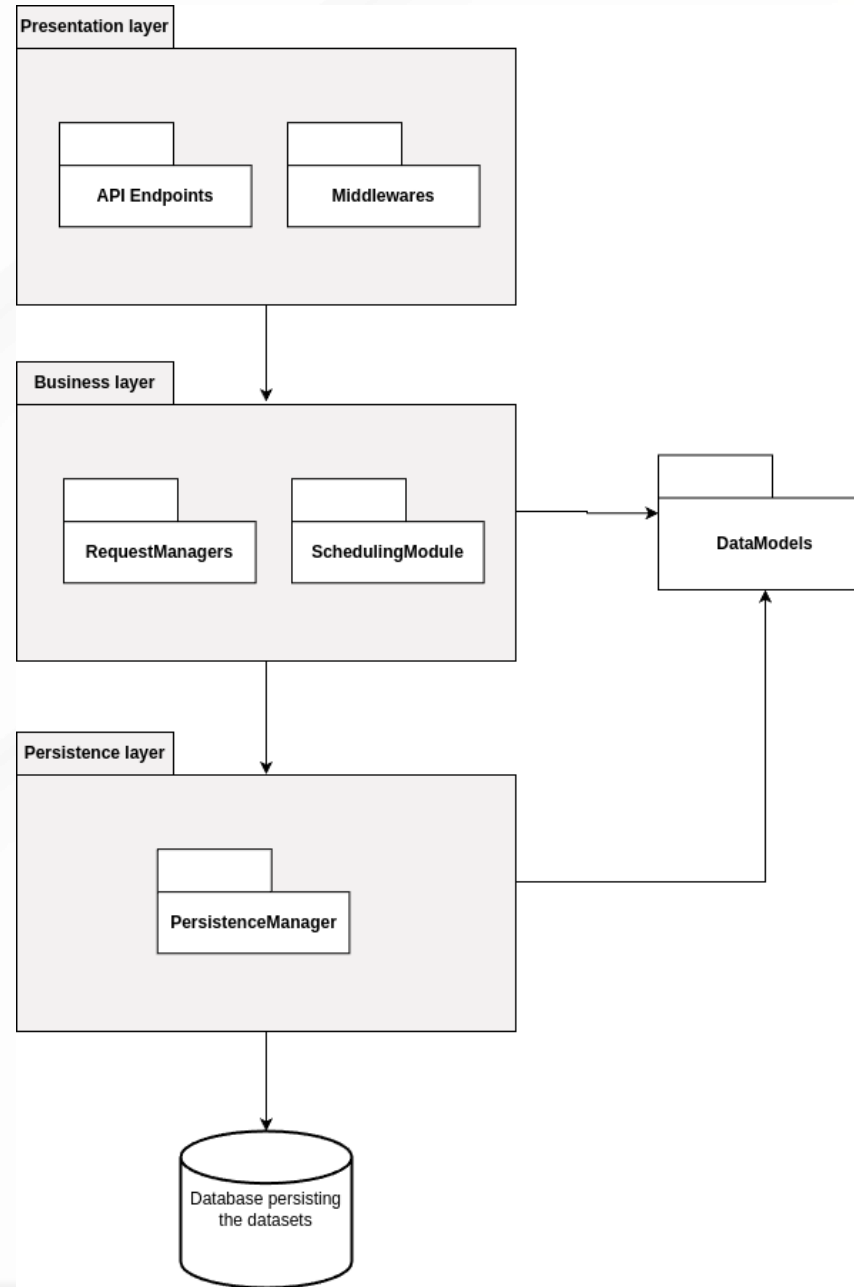
Appendix

Blackbox diagram



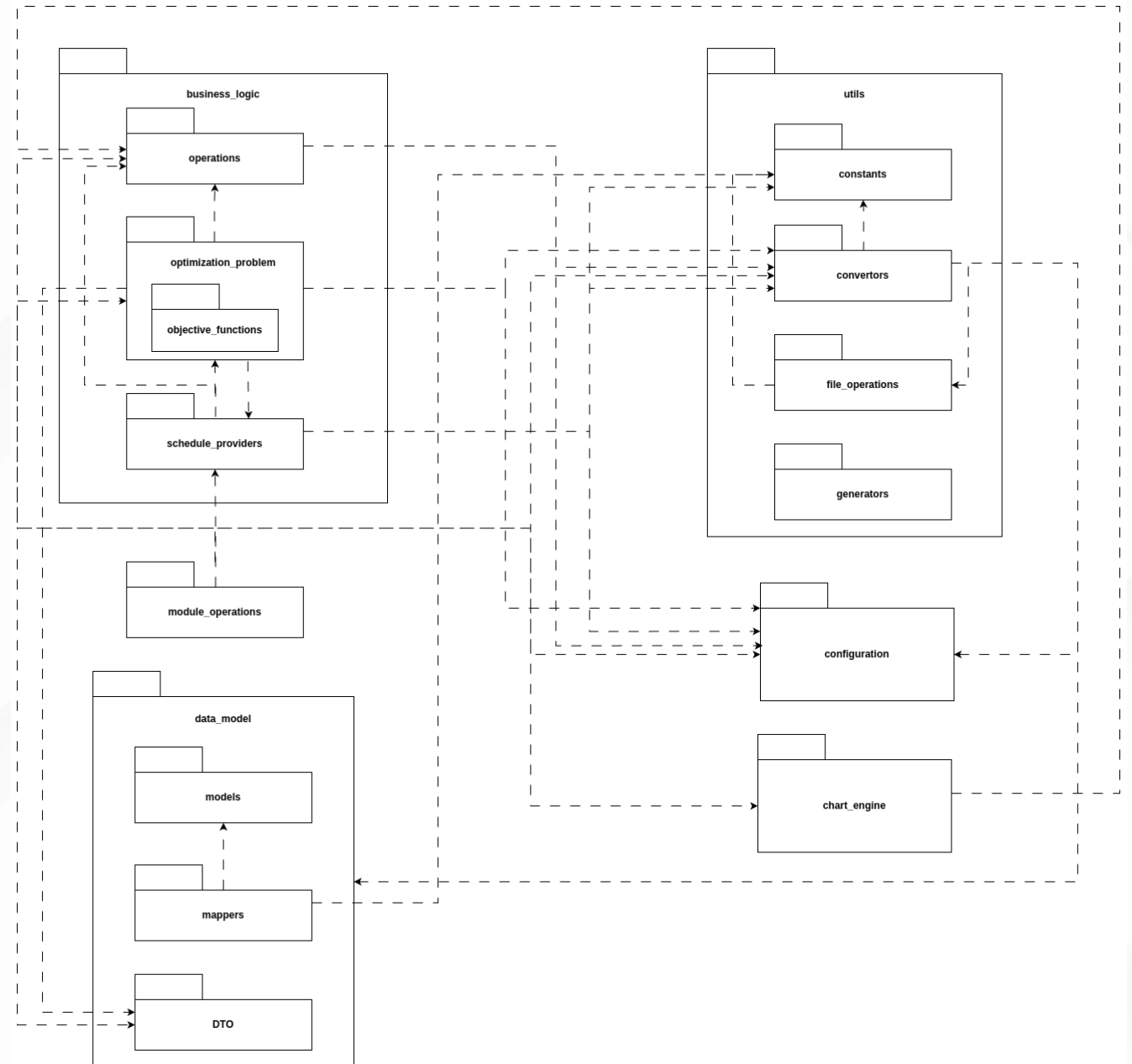
Appendix

Back-end package diagram



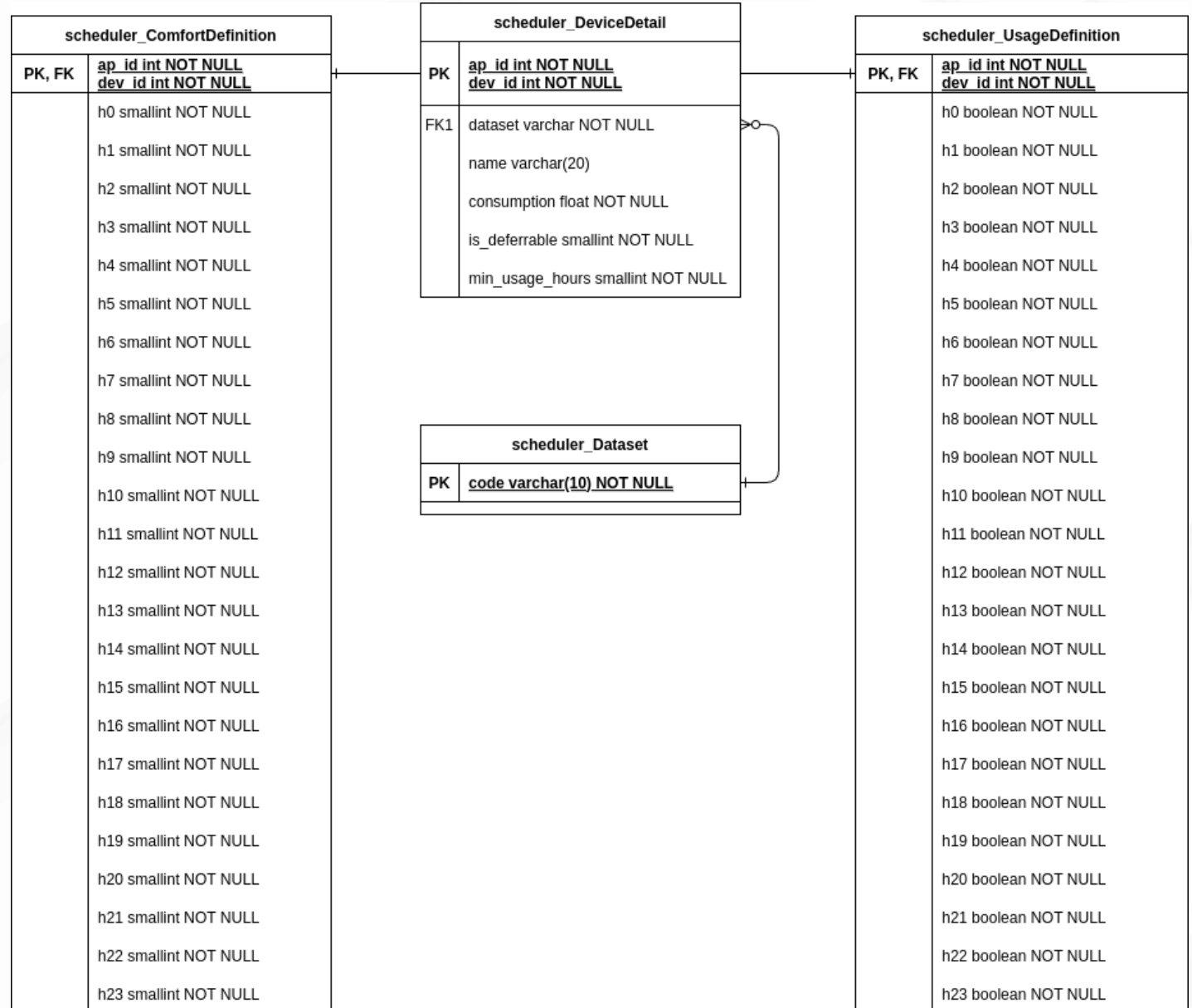
Appendix

Scheduling module package diagram



Appendix

Entity relationship diagram



Appendix

Deployment diagram

